TM Forum Specification

Resource Portability User Guide

TMF689

Maturity Level: General Availability (GA)	Team Approved Date: 16-Oct-2026
Release Status: Production	Approval status: Team Approved
Version: 5.0.0	IPR Mode: RAND

Notice

Copyright © TM Forum 2026. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the TM FORUM IPR Policy, must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an "AS IS" basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

TM FORUM invites any TM FORUM Member or any other party that believes it has patent claims that would necessarily be infringed by implementations of this TM Forum Standards Final Deliverable, to notify the TM FORUM Team Administrator and provide an indication of its willingness to grant patent licenses to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this deliverable.

TM FORUM invites any party to contact the TM FORUM Team Administrator if it is aware of a claim of ownership of any patent claims that would necessarily be infringed by implementations of this TM FORUM Standards Final Deliverable by a patent holder that is not willing to provide a license to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this TM FORUM Standards Final Deliverable. TM FORUM may include such claims on its website but disclaims any obligation to do so.

TM FORUM takes no position regarding the validity or scope of any intellectual property or other rights that might be claimed to pertain to the implementation or use of the technology described in this TM FORUM Standards Final Deliverable or the extent to which any license under such rights might or might not be available; neither does it represent that it has made any effort to identify any such rights. Information on TM FORUM's procedures

with respect to rights in any document or deliverable produced by a TM FORUM Collaboration Project Team can be found on the TM FORUM website. Copies of claims of rights made available for publication and any assurances of licenses to be made available, or the result of an attempt made to obtain a general license or permission for the use of such proprietary rights by implementers or users of this TM FORUM Standards Final Deliverable, can be obtained from the TM FORUM Team Administrator. TM FORUM makes no representation that any information or list of intellectual property rights will at any time be complete, or that any claims in such list are, in fact, Essential Claims.

Direct inquiries to the TM Forum office:

100 Enterprise Drive
Suite 301 #1649
Rockaway, NJ 070866, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

© TM Forum 2026. All Rights Reserved.

Table of Contents

Notice . . . i
Introduction . . . 1
Overview . . . 1
Objectives . . . 1
Relationship with Other TMF APIs . . . 1
Main Concepts and Resources . . . 2
PortabilityOrder . . . 2
PortabilityOrderItem . . . 3
CancelPortabilityOrder . . . 3
PortabilitySubject . . . 4
ServiceProvider . . . 4
Supporting Entities . . . 5
Portability Process Flow . . . 5
Notifications . . . 7
Sample Use Cases . . . 9
Overview of Portability Roles . . . 9
Use Case 1: Standard Mobile Number Portability (Recipient Perspective) . . . 9
Scenario . . . 9
Step 1: Create Portability Order (POST) . . . 9
Step 2: Query Portability Order Status (GET) . . . 10
Step 3: Update Portability Order (PATCH) . . . 10
Step 4: Event Notifications . . . 11
Step 5: Cancel Portability Order (Optional) . . . 12
Step 6: Query Cancellation Status . . . 13
Use Case 2: Donor Service Provider Processing . . . 13
Scenario . . . 13
Donor Activities . . . 13
Use Case 3: Multi-Level Donor Scenario . . . 15
Scenario . . . 15
Multi-Level Donor Tracking . . . 15
Multi-Level Coordination . . . 16
Use Case 4: Bulk Portability Order . . . 16
Scenario . . . 16

© TM Forum 2026. All Rights Reserved.

Bulk Order Structure . . . 17
Bulk Order Processing . . . 18
Business Rules and Constraints . . . 18
Support of polymorphism and extension patterns . . . 20
The @type Attribute . . . 20
The @referredType Attribute . . . 20
The @schemaLocation Attribute . . . 21
The @baseType Attribute . . . 21
RESOURCE MODEL . . . 22
Managed Entity and Task Resource Models . . . 22
PortabilityOrder resource . . . 22
CancelPortabilityOrder resource . . . 133
Notification Resource Models . . . 136
PortabilityOrder . . . 137
CancelPortabilityOrder . . . 143
API OPERATIONS . . . 146
Operations on PortabilityOrder . . . 146
Retrieves a portabilityOrder by ID . . . 146
List or find portabilityOrder objects . . . 149
Creates a portabilityOrder . . . 152
Updates partially a portabilityOrder . . . 158
Deletes a portabilityOrder . . . 173
Operations on CancelPortabilityOrder . . . 174
Retrieves a cancelPortabilityOrder by ID . . . 174
List or find cancelPortabilityOrder objects . . . 175
Creates a cancelPortabilityOrder . . . 176
Updates partially a cancelPortabilityOrder . . . 177
Deletes a cancelPortabilityOrder . . . 181
API NOTIFICATIONS . . . 183
Register listener . . . 183
Unregister listener . . . 184
Publish Event to listener . . . 184
Acknowledgements . . . 186
Release History . . . 186
Contributors to Document . . . 186

© TM Forum 2026. All Rights Reserved.

Introduction

The following document is the user guide of the REST API for Resource Portability management. It includes the model definition as well as all available operations.

Overview

The TMF689 Resource Portability API provides a standardized mechanism for managing the portability of resources (such as telephone numbers, network identifiers, or other service-related resources) between service providers. This API enables service providers to exchange resource portability information in a consistent and interoperable manner, supporting the business processes required for number portability and other resource transfer scenarios.

Objectives

The Resource Portability API serves the following key objectives:

- **Standardized Portability Process**: Provides a uniform interface for initiating, tracking, and managing resource portability orders across different service provider systems
- **Multi-Party Coordination**: Facilitates communication between donor service providers (losing the resource) and recipient service providers (gaining the resource)
- **Order Lifecycle Management**: Supports the complete lifecycle of portability orders from creation through completion or cancellation
- **Regulatory Compliance**: Enables compliance with regulatory requirements for resource portability in telecommunications markets
- **Automation Support**: Allows for automated processing of portability requests while maintaining visibility and control

Relationship with Other TMF APIs

The Resource Portability API integrates with several other TMF APIs to provide comprehensive service provider capabilities:

- **TMF639 Resource Inventory**: References resources that are subject to portability
- **TMF641 Service Ordering**: May be linked to service orders that trigger portability requirements
- **TMF632 Party Management**: References parties (customers, organizations) involved in the portability process

© TM Forum 2026. All Rights Reserved.

- **TMF669 Party Role**: ServiceProvider entities (donor and recipient) are modeled as PartyRole, enabling proper representation of service provider organizations and their role-specific attributes
- **TMF673 Geographic Address Management**: References service addresses associated with portability subjects
- **TMF634 Resource Catalog**: References resource specifications for resources being ported

Main Concepts and Resources

PortabilityOrder

The primary resource managed by this API. A `PortabilityOrder` represents a request to port one or more resources from a donor service provider to a recipient service provider. This is the central entity that orchestrates the entire portability process.

Portability-Specific Concepts:

- **Authorization and Compliance**: Portability orders require formal customer authorization before processing. The `portabilitySubject` entity captures authorization details including the authorization method, date, and supporting documentation. This ensures regulatory compliance and prevents unauthorized resource transfers.
- **Multi-Party Coordination**: The `donor` and `recipient` properties explicitly identify the two service providers involved in the portability transaction. This clear separation enables unambiguous responsibility assignment and facilitates inter-provider communication throughout the portability lifecycle.
- **Portability Request Tracking**: Industry-standard identifiers like `portRequestNumber` and `portRequestNumberVersion` enable tracking across provider systems and regulatory databases. The `groupRequestNumber` supports bulk portability scenarios where multiple resources are ported together as a coordinated batch.
- **Dispute Management**: The `dispute` flag indicates when a donor service provider challenges the portability request, triggering dispute resolution workflows. Disputes may arise from outstanding obligations, contract terms, or authorization issues.
- **Portability History**: The `orderRelationship` array maintains links to related portability orders, enabling tracking of multi-level donor scenarios where resources have been ported multiple times across different providers.

© TM Forum 2026. All Rights Reserved.

PortabilityOrderItem

Represents an individual resource within a portability order. Each order item tracks the portability of one specific resource.

Portability-Specific Concepts:

- **Activation Control**: The `portabilityActivationMode` determines whether the resource activation at the recipient provider occurs automatically at the scheduled time or requires manual intervention. Manual activation provides control for coordinated cutover scenarios.
- **Effective Date Management**: The `portabilityEffectiveDate` specifies the exact moment when the resource becomes active at the recipient provider and billing ceases at the donor provider. This date is critical for regulatory compliance and billing accuracy.
- **End User Association**: The `endUserName` identifies the actual user of the resource, which may differ from the account holder. This is important for validation and authorization in business scenarios where resources are assigned to employees or departments.
- **Processing Messages**: The `message` array contains portability-specific status updates, error codes, and inter-provider communications that track the technical progress of the portability action. These messages facilitate troubleshooting and provide audit trails for regulatory compliance.

CancelPortabilityOrder

A specialized resource for requesting cancellation of an existing portability order. This supports scenarios where a customer or service provider needs to abort a portability process that is already in progress.

Portability-Specific Concepts:

- **Cancellation Window**: Portability orders can typically be cancelled up until a specific point in the process (often before the donor provider begins deactivation). The `requestedCancellationDate` allows specification of when the cancellation should take effect, while `effectiveCancellationDate` records when it actually occurred.
- **Cancellation Reasons**: The `cancellationReason` captures why the portability is being aborted (e.g., customer changed mind, donor dispute, technical issues). This information is valuable for process improvement and regulatory reporting.
- **Order Reference**: The mandatory `portabilityOrder` reference links the cancellation request to the original portability order, maintaining the complete lifecycle history.

© TM Forum 2026. All Rights Reserved.

PortabilitySubject

Contains information about the customer or account holder associated with the portability request. This entity ensures proper customer identification and authorization tracking.

Portability-Specific Concepts:

- **Authorization Validation**: The portability subject captures detailed authorization information to validate that the customer has legitimately requested the portability. This includes the authorization method (electronic signature, verbal, written), authorization codes, and supporting documentation. Proper authorization is critical to prevent fraudulent portability requests.
- **Customer Identification**: Detailed name components (prefix, first name, middle initial, last name, suffix) and business name support accurate customer matching between donor and recipient provider systems. This reduces portability failures due to name mismatches.
- **Address Association**: The `address` property links the portability request to a specific location (service address, billing address, or other relevant address). This is particularly important for fixed-line portability where the resource is tied to a geographic location. This ensures the resource is ported to the correct address at the recipient provider.

ServiceProvider

A ServiceProvider is a PartyRole played by a Party (typically an Organization) that provides telecommunications or digital services. In portability orders, service providers are explicitly identified as either the donor or the recipient.

Portability-Specific Concepts:

- **Donor Service Provider**: The current provider losing the customer. The donor is responsible for validating the portability request, ensuring customer authorization is valid, checking for outstanding obligations, and releasing the resources at the scheduled time.
- **Recipient Service Provider**: The new provider gaining the customer. The recipient initiates the portability request, prepares systems to accept the ported resources, and activates services at the scheduled time.
- **PartyRole Structure**: As a PartyRole, ServiceProvider includes an `engagedParty` reference to the actual Organization providing the services, along with role-specific characteristics, contact information, and agreements.

© TM Forum 2026. All Rights Reserved.

Supporting Entities

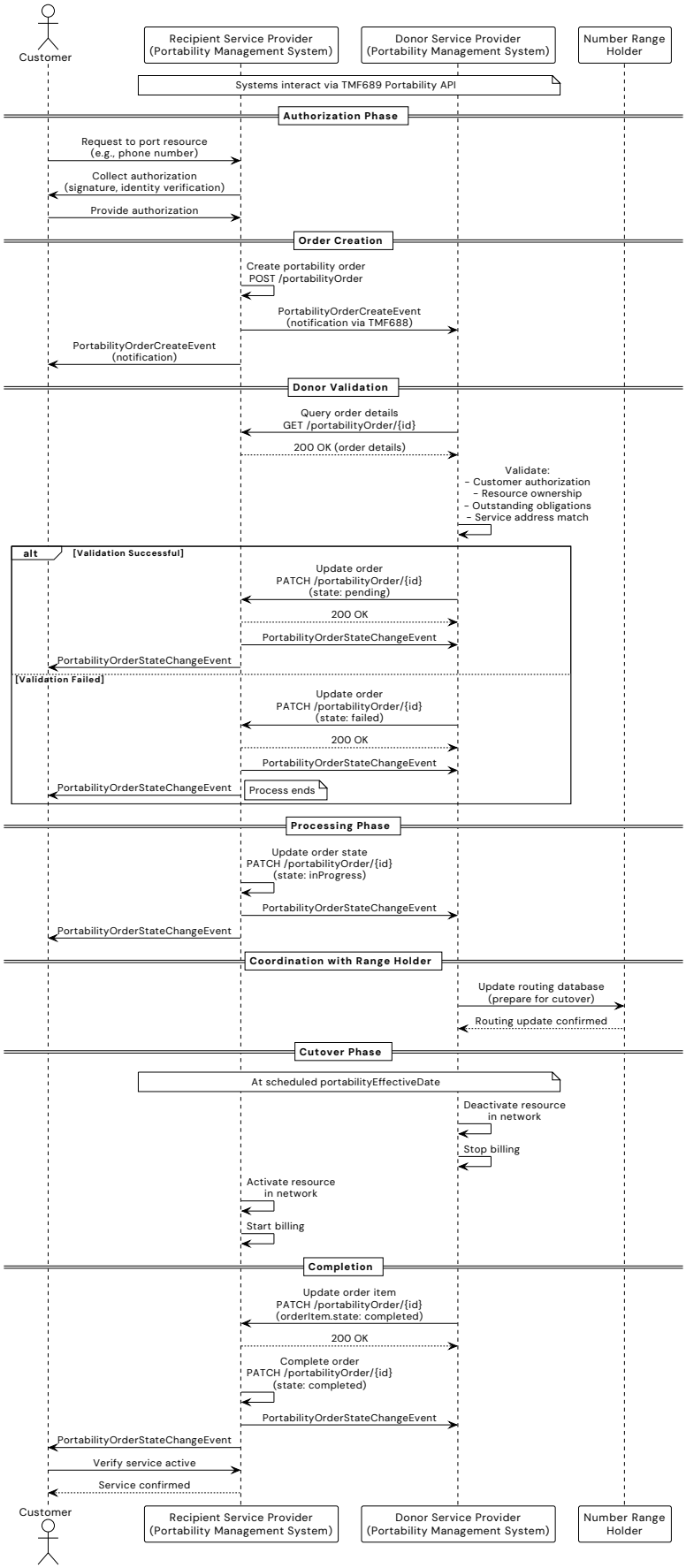
OrderRelationship: Links related portability orders to other orders (e.g., linking a cancellation to the original order, linking orders in a multi-level donor chain, or referencing service orders that triggered the portability)

ErrorMessage: Carries processing messages, error codes, and status information throughout the portability lifecycle. Used in the `message` array of order items to track technical progress and facilitate troubleshooting.

Portability Process Flow

The following sequence diagram illustrates the key actors and interactions in a successful resource portability process:

© TM Forum 2026. All Rights Reserved.



© TM Forum 2026. All Rights Reserved.

Key Actors:

- **Customer**: The end user requesting the portability
- **Recipient Service Provider (Portability Management System)**: The new provider gaining the customer. Their portability management system initiates the portability order and exposes the TMF689 API for coordination with the donor
- **Donor Service Provider (Portability Management System)**: The current provider losing the customer. Their portability management system validates the request, releases the resource, and consumes the TMF689 API to interact with the recipient's order
- **Number Range Holder**: Authority maintaining the routing database (for number portability scenarios)

Note: The TMF689 Portability API is the standardized interface used by the service providers' portability management systems to exchange information. The API is exposed by components under the responsibility of the organizations (service providers), not as a standalone entity.

Critical Success Factors:

- **Customer Authorization**: Must be validated by donor before proceeding
- **Coordination**: Both providers must update their systems at the scheduled time
- **Event Notifications**: Keep all parties informed of progress and state changes
- **Timing Precision**: Cutover must occur at the exact portabilityEffectiveDate to minimize service disruption

Notifications

The API supports comprehensive event notifications to keep all parties informed of portability order lifecycle changes, including:

PortabilityOrder Events:

- **Create Events** (`PortabilityOrderCreateEvent`): Notification when a new portability order is created
- **State Change Events** (`PortabilityOrderStateChangeEvent`): Notification when the order state transitions (e.g., acknowledged → inProgress → completed). This is the primary event for tracking order lifecycle and covers scenarios previously handled by separate cancellation or completion events.
- **Attribute Value Change Events** (

© TM Forum 2026. All Rights Reserved.

`PortabilityOrderAttributeValueChangeEvent`): Notification when order attributes are modified (excluding state changes)

- **Information Required Events** (`PortabilityOrderInformationRequiredEvent`): Notification when additional information is needed to process the order (e.g., missing authorization, incomplete customer details)

CancelPortabilityOrder Events:

- **Create Events** (`CancelPortabilityOrderCreateEvent`): Notification when a cancellation request is created
- **State Change Events** (`CancelPortabilityOrderStateChangeEvent`): Notification when the cancellation request state transitions
- **Information Required Events** (`CancelPortabilityOrderInformationRequiredEvent`): Notification when additional information is needed to process the cancellation

These notifications enable real-time coordination between donor and recipient service providers, customer systems, and regulatory reporting systems.

© TM Forum 2026. All Rights Reserved.

Sample Use Cases

This section describes the key use cases for resource portability, illustrating the different roles and scenarios involved in the portability process.

Overview of Portability Roles

In resource portability scenarios, there are typically three key roles:

- **Recipient Service Provider**: The service provider that will gain the resource (the customer's new provider)
- **Donor Service Provider**: The service provider that currently owns the resource (the customer's old provider)
- **Multi-Level Donor**: In complex scenarios, a resource may have been ported multiple times, creating a chain of previous donors

Use Case 1: Standard Mobile Number Portability (Recipient Perspective)

This use case demonstrates the complete lifecycle of porting a mobile telephone number from one service provider (donor) to another (recipient), from the perspective of the recipient service provider.

Scenario

Rahul K Sharma is a customer of Notel Communications and wants to switch to NewTel Services while keeping his mobile number +61412345678. The portability process involves coordination between the two service providers through the Resource Portability API.

Step 1: Create Portability Order (POST)

The recipient service provider (NewTel Services) creates a portability order to initiate the porting process.

Request: `POST /portabilityOrder`

The request includes: * Customer authorization date and details * Portability subject information (customer name, service address) * Donor and recipient service provider information * Order item containing the resource to be ported (telephone number) * Requested dates for the portability to take effect * Priority level (high, medium, low)

Response: `201 Created`

© TM Forum 2026. All Rights Reserved.

The system creates the portability order with an initial state of "acknowledged" and assigns a unique order ID (e.g., "999"). The response includes the complete order with server-generated fields like `id`, `href`, `creationDate`, and initial state.

Example: See operation example `Creates a PortabilityOrder` in Part 5 (Operations) of this specification.

Step 2: Query Portability Order Status (GET)

Either service provider or the customer system can query the status of portability orders.

Request: `GET /portabilityOrder?state=inProgress&fields=@type,state,id,href`

This retrieves all portability orders in "inProgress" state, returning only the specified fields to optimize the response.

Response: `200 OK`

Returns a collection of portability orders matching the query criteria.

Example: See operation example `portabilityOrder_GET_200_12` in Part 5 (Operations) of this specification.

Request: `GET /portabilityOrder/AQAWI`

Retrieves the complete details of a specific portability order by its ID.

Response: `200 OK`

Returns the full portability order resource with all nested details.

Example: See operation example `portabilityOrder_YDJTH_GET_200_14` in Part 5 (Operations) of this specification.

Step 3: Update Portability Order (PATCH)

As the portability process progresses, the order state and attributes are updated using PATCH operations. The API supports three PATCH formats:

Option A: Merge Patch (`application/merge-patch+json`)

Request: `PATCH /portabilityOrder/AQAWI`

Updates the order state to "inProgress" and adds processing notes using a simple merge approach where specified fields replace existing values.

© TM Forum 2026. All Rights Reserved.

Response: `200 OK`

Returns the complete updated portability order.

Example: See operation example `portabilityOrder_AQAWI_PATCH_200_application_merge-patch+json_16` in Part 5 (Operations) of this specification.

Option B: JSON Patch (`application/json-patch+json`)

Request: `PATCH /portabilityOrder/MGEJD`

Uses JSON Patch operations (add, replace, remove) to make precise modifications to the order, such as updating the expected completion date.

Response: `200 OK`

Returns the complete updated portability order.

Example: See operation example `portabilityOrder_MGEJD_PATCH_200_application_json-patch+json_17` in Part 5 (Operations) of this specification.

Option C: JSON Patch Query (`application/json-patch-query+json`)

Request: `PATCH /portabilityOrder/FHAGB`

Uses JSON Patch Query format to update the order to a "failed" state with detailed error information.

Response: `200 OK`

Returns the complete updated portability order.

Example: See operation example `portabilityOrder_FHAGB_PATCH_200_application_json-patch-query+json_18` in Part 5 (Operations) of this specification.

Step 4: Event Notifications

Throughout the portability lifecycle, the system sends event notifications to registered listeners:

Create Event: When the portability order is first created

The notification payload contains the entire portability order entity.

© TM Forum 2026. All Rights Reserved.

Example: See operation example `listener_portabilityOrderCreateEvent_POST_204_11` in Part 5 (Operations) of this specification.

Attribute Value Change Event: When order attributes are modified

The notification payload contains only the `@type`, `id`, `href`, and the changed attributes in merge-patch format.

Example: See operation example `listener_portabilityOrderAttributeValueChangeEvent_POST_204_8` in Part 5 (Operations) of this specification.

State Change Event: When the order state transitions (e.g., from "inProgress" to "completed")

The notification payload contains only the `@type`, `id`, `href`, and the new state value.

Example: State change events use the generic attribute value change event structure shown above.

Ported Service Event: When a resource has been successfully ported

Notifies all parties that the portability has been completed for a specific service.

Example: See operation example `listener_portabilityOrderPortedServiceEvent_POST_204_12` in Part 5 (Operations) of this specification.

Step 5: Cancel Portability Order (Optional)

If the customer or service provider needs to cancel the portability before completion:

Request: `POST /cancelPortabilityOrder`

Creates a cancellation request for an existing portability order, specifying the reason and requested cancellation date.

Response: `201 Created` or `202 Accepted`

The system creates a cancel order record and initiates the cancellation process.

Examples: * See operation example `cancelPortabilityOrder_POST_201_2` in Part 5 (Operations) of this specification. * See operation example `cancelPortabilityOrder_POST_202_1` in Part 5 (Operations) of this specification.

© TM Forum 2026. All Rights Reserved.

Cancel Order Notification: See operation example `listener_cancelPortabilityOrderCreateEvent_POST_204_5` in Part 5 (Operations) of this specification.

Step 6: Query Cancellation Status

Request: `GET /cancelPortabilityOrder`

Retrieves all cancellation orders.

Response: `200 OK`

Returns a collection of cancel portability orders.

Example: See operation example `cancelPortabilityOrder_GET_200_1` in Part 5 (Operations) of this specification.

Request: `GET /cancelPortabilityOrder/CPO-001`

Retrieves a specific cancellation order by ID.

Response: `200 OK`

Returns the complete cancel portability order resource.

Example: See operation example `cancelPortabilityOrder_YDDWJ_GET_200_3` in Part 5 (Operations) of this specification.

Use Case 2: Donor Service Provider Processing

This use case illustrates the portability process from the perspective of the donor service provider (the provider losing the customer).

Scenario

Notel Communications (donor) receives a portability request from NewTel Services (recipient) to port customer Rahul K Sharma's mobile number +61412345678. Notel must validate the request and coordinate the resource release.

Donor Activities

Step 1: Receive Portability Notification

Notel receives a notification (via the API or through inter-provider messaging) that a portability order has been created for one of their resources.

© TM Forum 2026. All Rights Reserved.

Step 2: Query and Validate the Order

Request: `GET /portabilityOrder/999`

Notel retrieves the portability order details to validate: * Customer authorization is valid * Resource (phone number) is active and belongs to the specified customer * No outstanding obligations (unpaid bills, contract terms) * Service address and customer details match their records

Step 3: Update Order Status

If validation succeeds, Notel updates the order to indicate donor acceptance:

Request: `PATCH /portabilityOrder/999`

```
{
  "state": "pending",
  "note": [{
    "id": "2",
    "author": "Notel Portability Team",
    "date": "2026-03-20T14:30:00.000Z",
    "text": "Donor validation completed. Resource ready for porting."
  }]
}
```

If validation fails (e.g., customer mismatch, outstanding obligations), Notel updates the order to "failed" state with appropriate reason codes.

Step 4: Execute Resource Release

When the portability effective date arrives, Notel: * Deactivates the resource in their network * Updates billing systems to stop charging the customer * Confirms completion via PATCH operation

Request: `PATCH /portabilityOrder/999`

```
{
  "state": "completed",
  "completionDate": "2026-03-25T10:00:00.000Z",
  "orderItem": [{
    "id": "1",
    "state": "completed",
    "portabilityEffectiveDate": "2026-03-25T10:00:00.000Z"
  }]
}
```

© TM Forum 2026. All Rights Reserved.

Use Case 3: Multi-Level Donor Scenario

This use case demonstrates a complex scenario where a resource has been ported multiple times, creating a chain of donor service providers.

Scenario

Customer Priya Mehta's mobile number +61498765432 has a portability history: * **Original Provider**: TelecomOne (2020-2023) * **First Port**: To MobileNet (2023-2025) - TelecomOne became the donor * **Second Port**: To ConnectPlus (2025-present) - MobileNet became the donor * **Current Port Request**: To UltraTel (2026) - ConnectPlus is now the donor

Multi-Level Donor Tracking

The portability order includes relationship information to track the portability history:

PortabilityOrder Structure:


```
{
  "@type": "PortabilityOrder",
  "id": "PO-2026-001",
  "donor": {
    "@type": "ServiceProvider",
    "id": "SP-DONOR",
    "name": "ConnectPlus Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-CP-
001",
      "@referredType": "Organization",
      "id": "ORG-CP-001",
      "name": "ConnectPlus Pty Ltd"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "id": "SP-RECIPIENT",
    "name": "UltraTel Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-UT-
001",
      "@referredType": "Organization",
      "id": "ORG-UT-001",
      "name": "UltraTel Communications"
    }
  },
  "orderItem": [{
    "id": "1",
    "resource": {
      "@type": "LogicalResource",
```


© TM Forum 2026. All Rights Reserved.

```
      "value": "+61498765432",
      "resourceSpecification": {
        "@type": "ResourceSpecificationRef",
        "@referredType": "ResourceSpecification",
        "id": "MSISDN-SPEC",
        "name": "Mobile Number"
      },
      "note": [{
        "id": "1",
        "text": "Previous portability history: TelecomOne (original) ¬ MobileNet (2023)
¬ ConnectPlus (2025)",
        "author": "Portability System",
        "date": "2026-03-25T09:00:00.000Z"
      }]
    }
  }]
}
```

Multi-Level Coordination

In multi-level donor scenarios:

1. **Current Donor (ConnectPlus)** validates the request against their records
2. **Number Range Holder** (may be TelecomOne, the original provider) maintains the authoritative routing database
3. **All Previous Donors** may need to update their historical records for regulatory compliance
4. **Recipient (UltraTel)** coordinates with the current donor and potentially the range holder

The API supports this through: * **Order Relationships**: Link related portability orders across the history * **Notes**: Document the portability chain for audit and troubleshooting * **Related Parties**: Reference all organizations involved in the portability history * **Resource Relationships**: Track the resource's movement through different providers

Use Case 4: Bulk Portability Order

This use case demonstrates porting multiple resources in a single order, common when a business customer switches providers.

Scenario

ABC Corporation is moving from OldTel to NewCom and needs to port 50 telephone numbers simultaneously.

© TM Forum 2026. All Rights Reserved.

Bulk Order Structure

The portability order contains multiple order items, each representing one resource:

Request: `POST /portabilityOrder`

```
{
  "@type": "PortabilityOrder",
  "priority": "high",
  "requestedCompletionDate": "2026-03-30T18:00:00.000Z",
  "groupRequestNumber": "GRP-2026-ABC-001",
  "portabilitySubject": {
    "@type": "PortabilitySubject",
    "name": "ABC Corporation",
    "businessName": "ABC Corporation Pty Ltd",
    "authorization": {
      "name": "Bulk Portability Authorization",
      "givenDate": "2026-03-20T10:00:00.000Z",
      "state": "approved",
      "signatureRepresentation": "digital",
      "approver": {
        "@type": "RelatedParty",
        "@referredType": "Individual",
        "id": "AUTH-001",
        "name": "Jane Smith",
        "role": "Authorized Representative"
      }
    },
    "contactMedium": [...]
  },
  "donor": {
    "@type": "ServiceProvider",
    "name": "OldTel Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-
OLDTEL-001",
      "id": "ORG-OLDTEL-001",
      "@referredType": "Organization",
      "name": "OldTel Communications"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "name": "NewCom Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-
NEWCOM-001",
      "id": "ORG-NEWCOM-001",
      "@referredType": "Organization",
      "name": "NewCom Services"
    }
  },
  "orderItem": [
```

© TM Forum 2026. All Rights Reserved.

```
    {
      "id": "1",
      "action": "port",
      "resource": {
        "@type": "LogicalResource",
        "value": "+61412345001",
        "resourceSpecification": {...}
      }
    },
    {
      "id": "2",
      "action": "port",
      "resource": {
        "@type": "LogicalResource",
        "value": "+61412345002",
        "resourceSpecification": {...}
      }
    },
    // ... items 3-50
  ]
}
```

Bulk Order Processing

- **Group Request Number**: Links all resources in the bulk order for coordinated processing
- **Individual Item States**: Each order item can progress independently (some may complete while others are pending)
- **Partial Completion**: The overall order state reflects the aggregate state of all items
- **Rollback Handling**: If critical items fail, the order may support cancellation of remaining items

Business Rules and Constraints

- **Customer Authorization**: Portability orders require formal customer authorization before processing can begin. The `portabilitySubject.authorization` entity captures comprehensive authorization details including:
  - **Authorization Method**: How the customer provided authorization (e.g., electronic signature, written consent, verbal authorization)
  - **Authorization Date**: When the customer authorized the portability request (`givenDate`)
  - **Signature**: Digital or scanned physical signature as an attachment
  - **Approver**: The party responsible for validating the authorization (typically the recipient service provider or an authorized representative)

© TM Forum 2026. All Rights Reserved.

- **Authorization State**: Whether the authorization has been approved or declined

```
This structured authorization model ensures regulatory compliance and prevents
fraudulent portability requests. The donor service provider validates the
authorization details before releasing the resource.
```

- **Service Provider Roles**: Each order must identify both donor and recipient service providers with their respective roles. The donor is responsible for validating authorization and releasing resources, while the recipient initiates the request and prepares systems to accept the ported resources.
- **Timing**: Orders include requested, expected, and actual completion dates to manage customer expectations and coordinate the cutover between providers.
- **State Transitions**: Orders follow a defined state machine (acknowledged → pending → inProgress → completed/failed/cancelled) with state changes triggering notifications to all involved parties.
- **Resource Identification**: Resources being ported must be uniquely identified with proper references to resource inventory, including resource specifications and characteristics.
- **Cancellation Window**: Cancellations may be requested at any point before completion, subject to business rules and regulatory requirements. The cancellation window typically closes when the donor begins resource deactivation.
- **Multi-Level History**: Systems should maintain portability history for audit, troubleshooting, and regulatory compliance, tracking the chain of previous donors when resources have been ported multiple times.
- **Bulk Processing**: Multiple resources can be ported in a single order with coordinated or independent item processing, using the `groupRequestNumber` to link related portability requests.

© TM Forum 2026. All Rights Reserved.

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities. In the Resource Portability API, this is demonstrated by `LogicalResource` (representing ported resources like telephone numbers) inheriting from the base `Resource` entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type Attribute

The `@type` attribute provides a way to represent the actual class type of an entity. For example, within a portability order, the `resource` property may contain different types of resources being ported. The `@type` attribute indicates the specific resource type:

- A telephone number resource might have `@type: "LogicalResource"` or a more specific subclass like `"TelephoneNumber"`
- A portability order would have `@type: "PortabilityOrder"`

All resources and sub-resources of this API have `@type` attributes that can be provided when this is useful.

The @referredType Attribute

The `@referredType` attribute is used within reference entities to explicitly denote the actual entity type of the referred class. For example:

- In a `PortabilityOrderRef` object referencing a portability order, `@referredType` would be `"PortabilityOrder"`
- In a `ResourceRef` object referencing a logical resource being ported, `@referredType` would be `"LogicalResource"` or the specific resource subclass

Notice that in reference entities the `@type`, when used, denotes the class type of the reference itself (such as `PortabilityOrderRef`), and not the class type of the referred object. However, since reference classes are rarely sub-classed, `@type` is generally not useful in reference objects.

© TM Forum 2026. All Rights Reserved.

The @schemaLocation Attribute

The `@schemaLocation` property can be used in resources to allow specifying user-defined properties of an entity or to specify the expected characteristics of an entity. This is particularly useful when:

- Extending the `PortabilityOrder` entity with provider-specific attributes
- Adding custom fields to `PortabilityOrderItem` for regional regulatory requirements
- Defining specialized resource types beyond the standard `LogicalResource`

The @baseType Attribute

The `@baseType` attribute gives a way to provide explicitly the base class of a given resource that has been extended. For example:

- A `LogicalResource` instance would have `@baseType: "Resource"`
- A custom extended resource type like `TelephoneNumber` would have `@baseType: "LogicalResource"`

Note: `@baseType` should only be populated when there is a functionally significant base class (true inheritance). Do not populate `@baseType` with the same value as `@type`, or with technical classes such as `Entity`, `EntityRef`, `Extensible`, or `Addressable`.

© TM Forum 2026. All Rights Reserved.

RESOURCE MODEL

Managed Entity and Task Resource Models

PortabilityOrder resource

A Portability Order represents a formal request to transfer resources (such as telephone numbers, services, or other portable assets) from one service provider (donor) to another service provider (recipient), while maintaining service continuity for the end user. This entity manages the complete lifecycle of the portability process, including authorization, scheduling, execution, and completion tracking.

Resource model

© TM Forum 2026. All Rights Reserved.

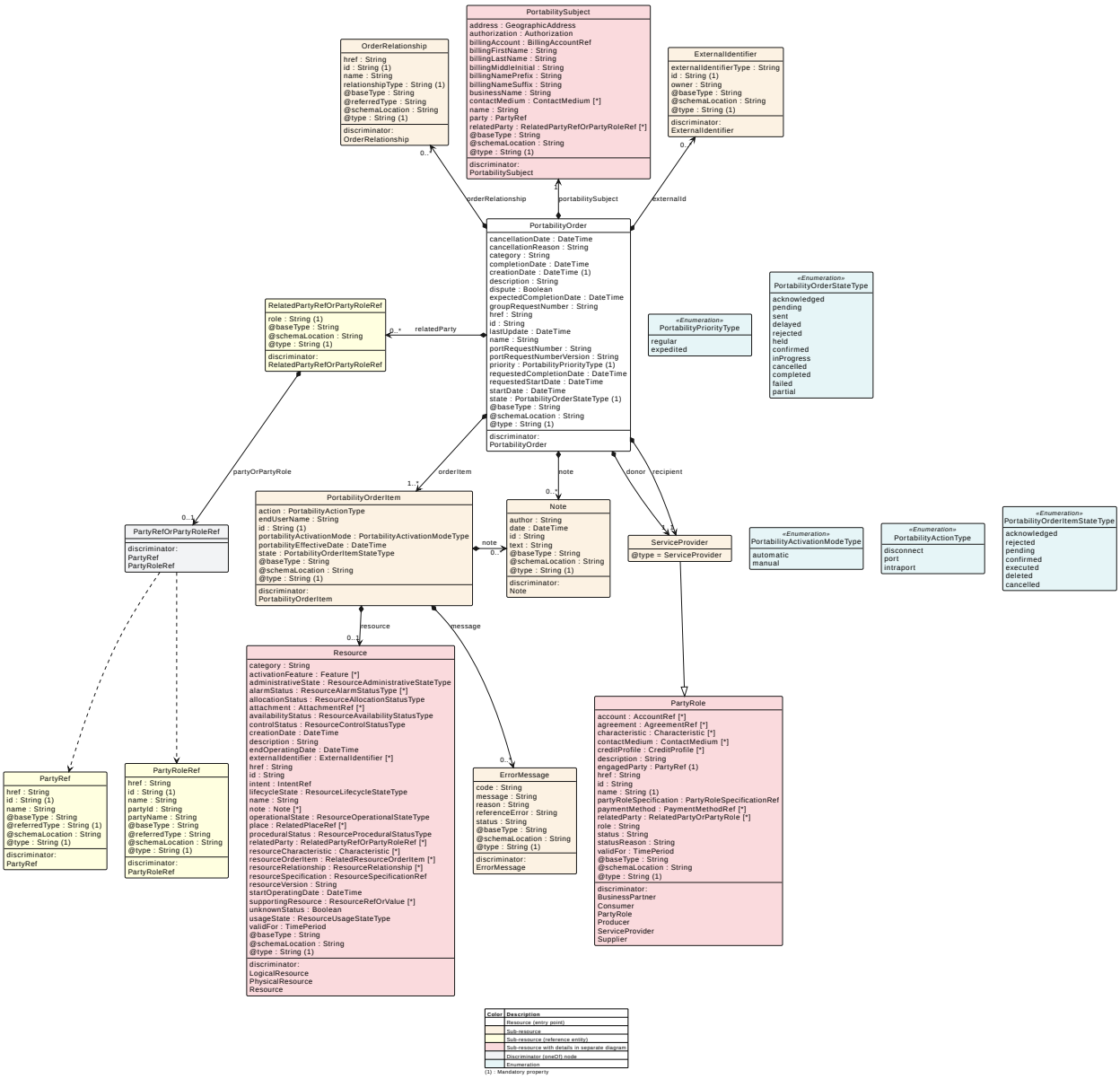


Figure 1 - PortabilityOrder

© TM Forum 2026. All Rights Reserved.

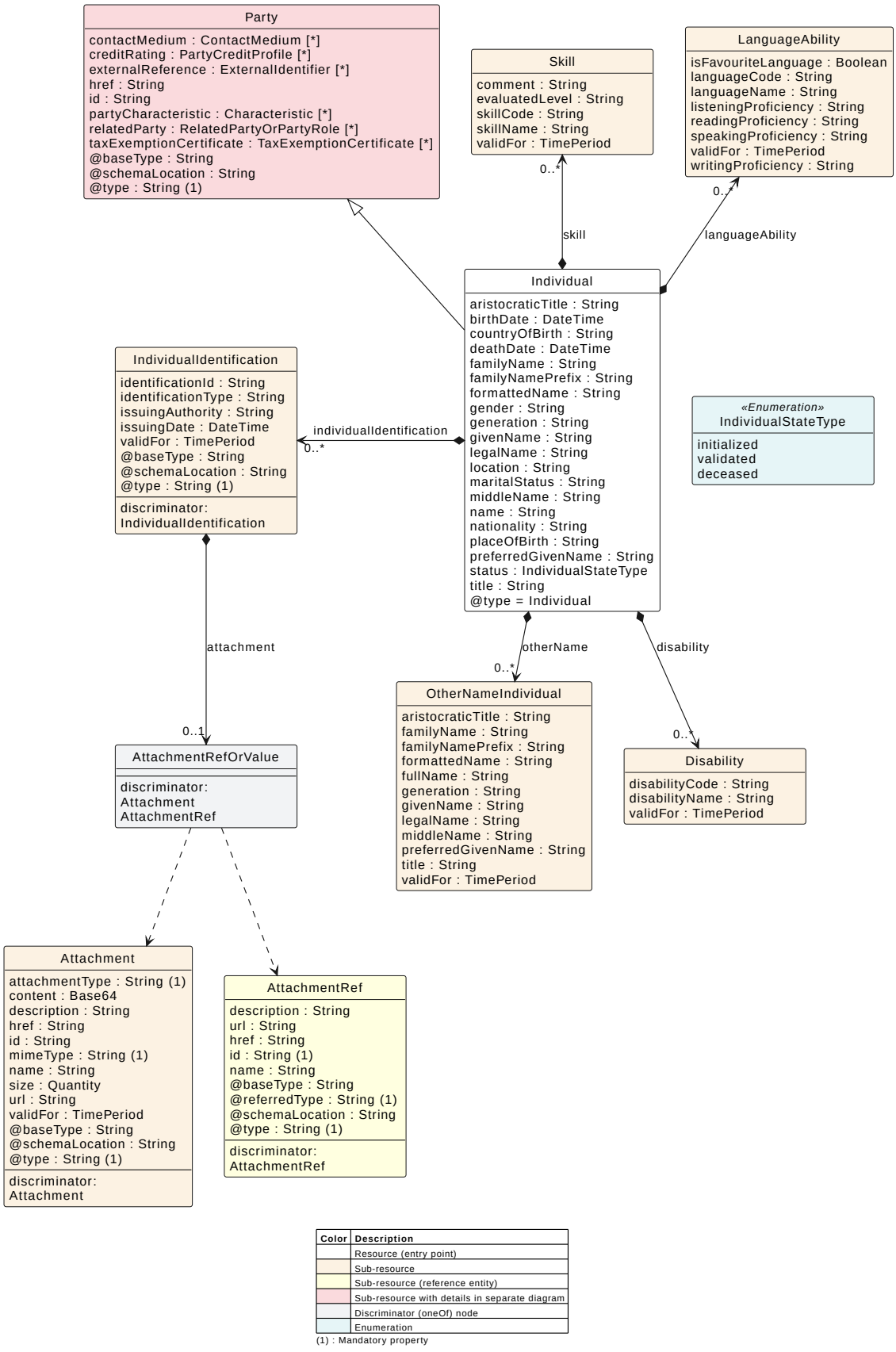


Figure 2 - Individual

© TM Forum 2026. All Rights Reserved.

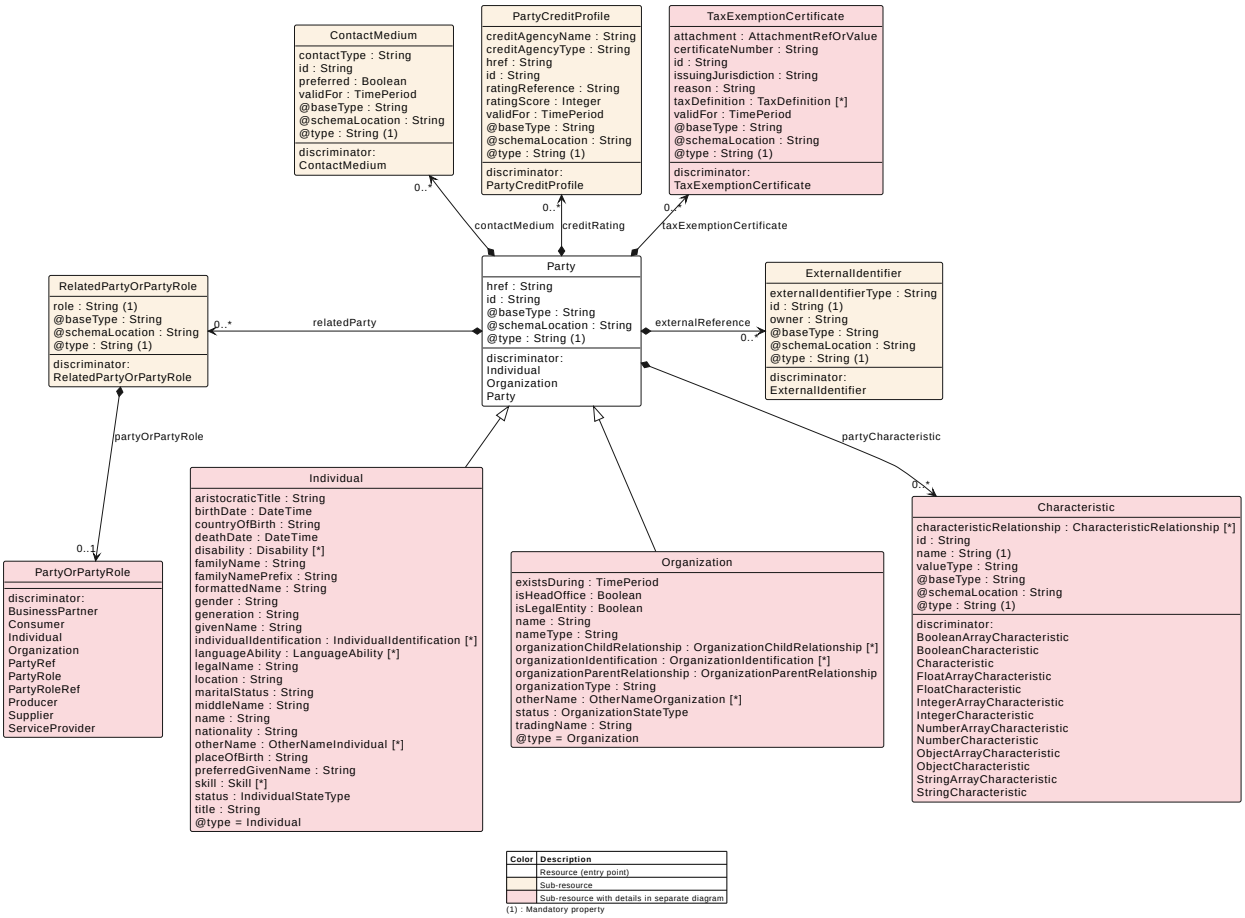


Figure 3 - Party

© TM Forum 2026. All Rights Reserved.

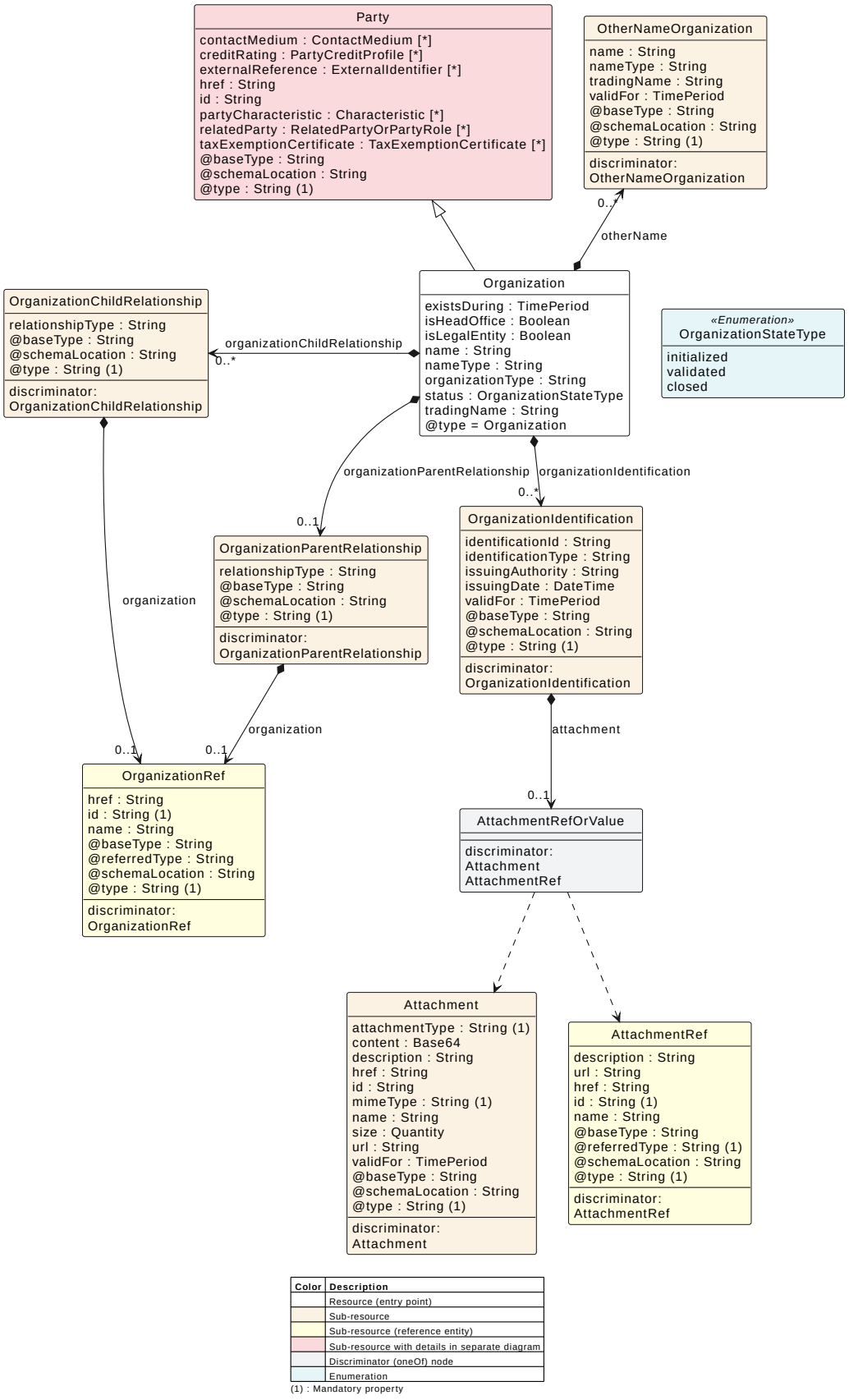


Figure 4 - Organization

© TM Forum 2026. All Rights Reserved.

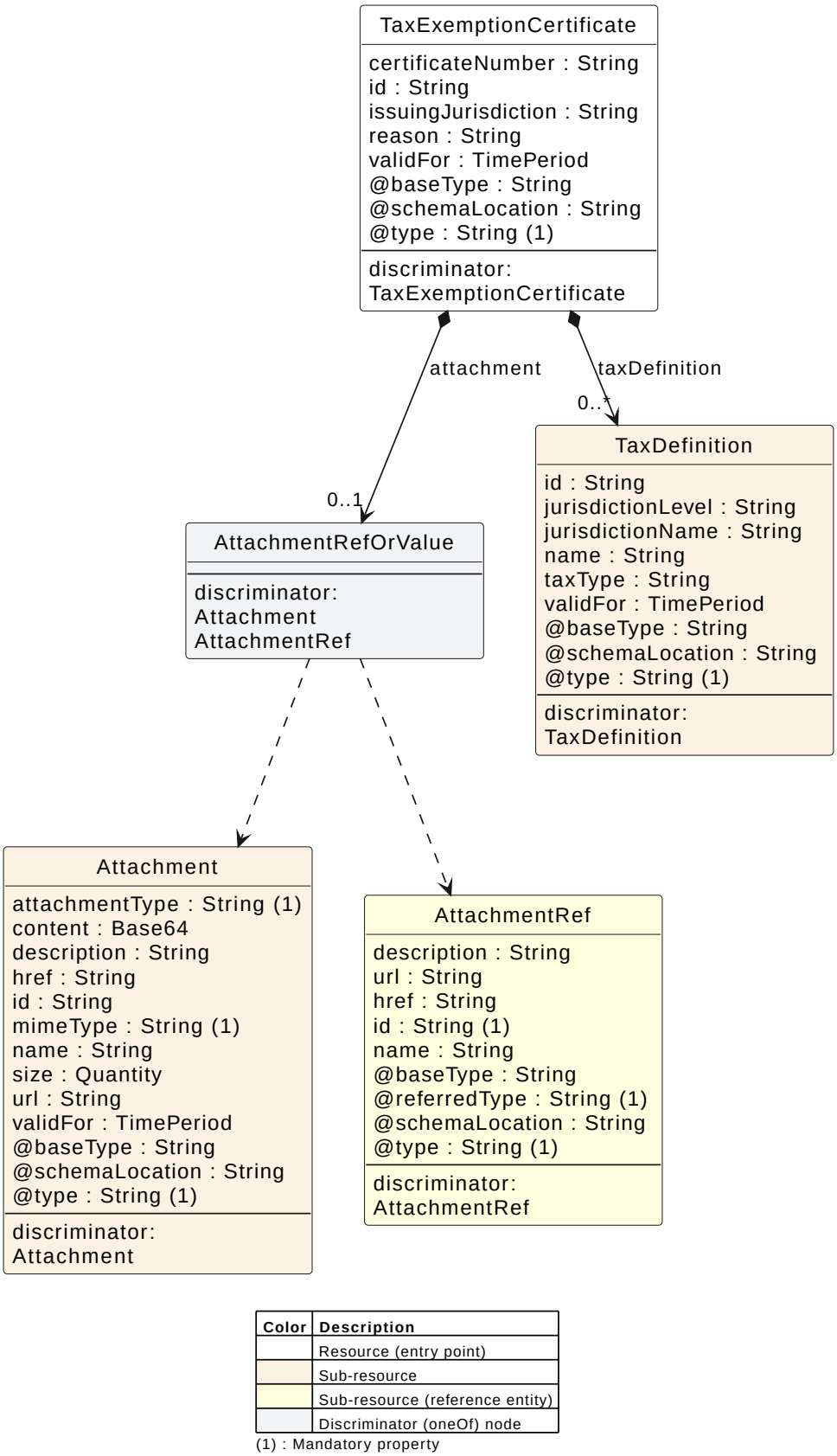


Figure 5 - TaxExemptionCertificate

© TM Forum 2026. All Rights Reserved.

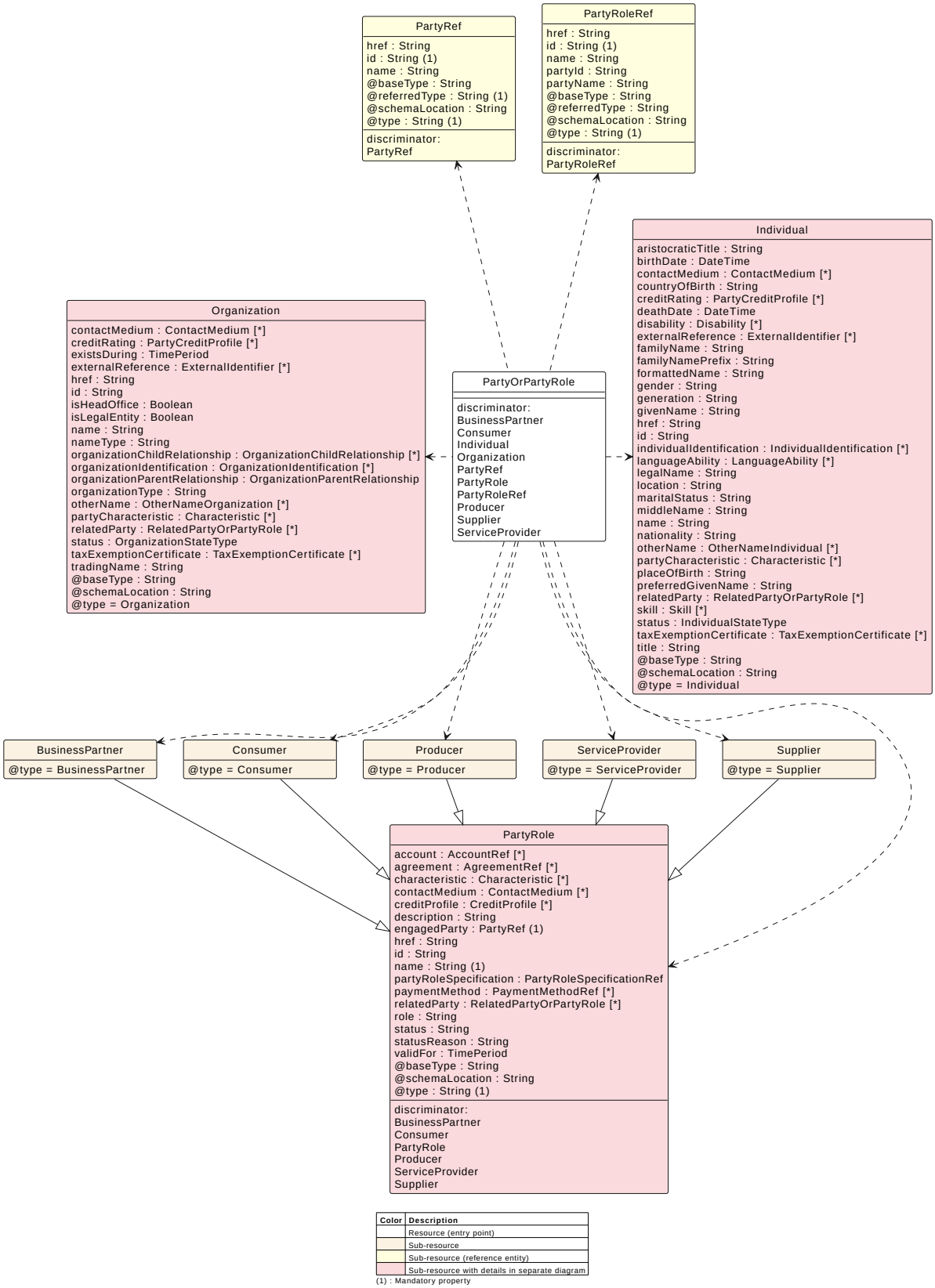


Figure 6 - PartyOrPartyRole

© TM Forum 2026. All Rights Reserved.

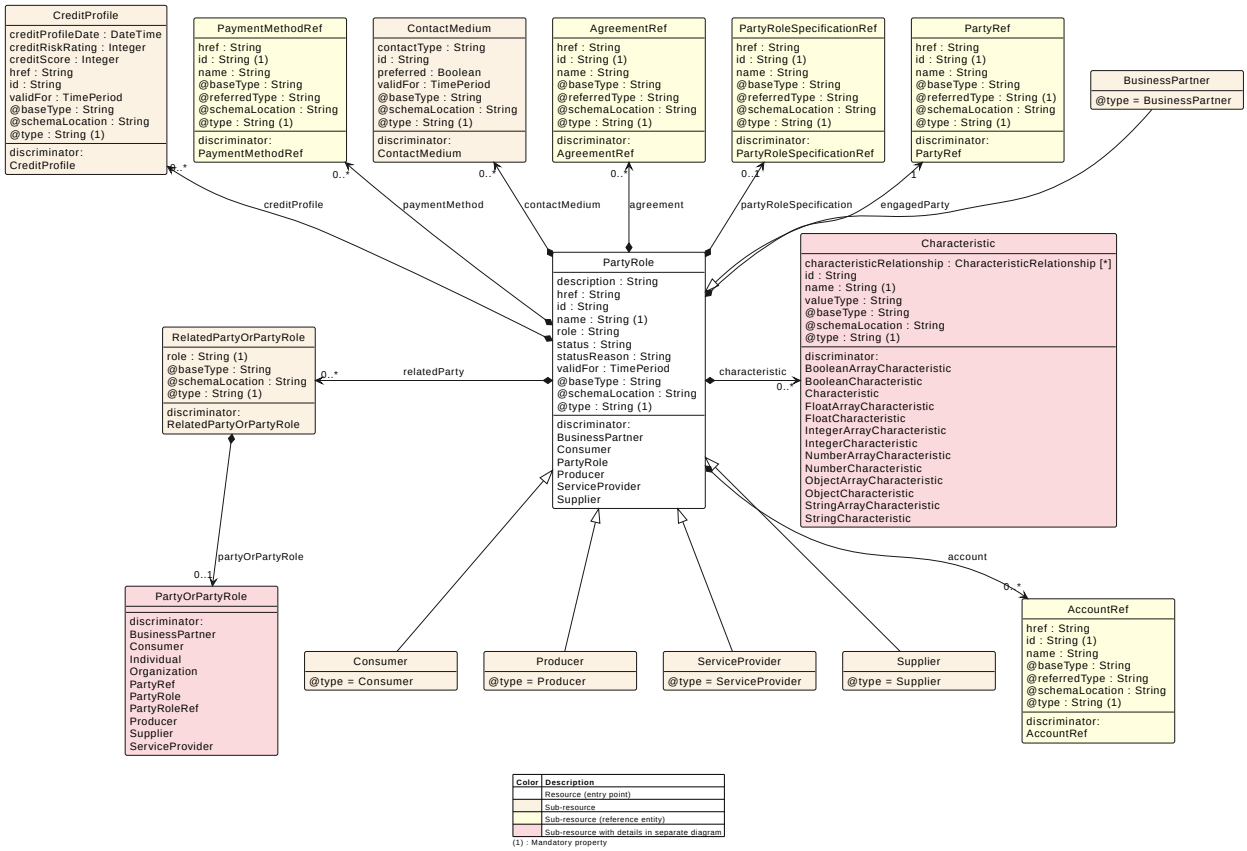


Figure 7 - PartyRole

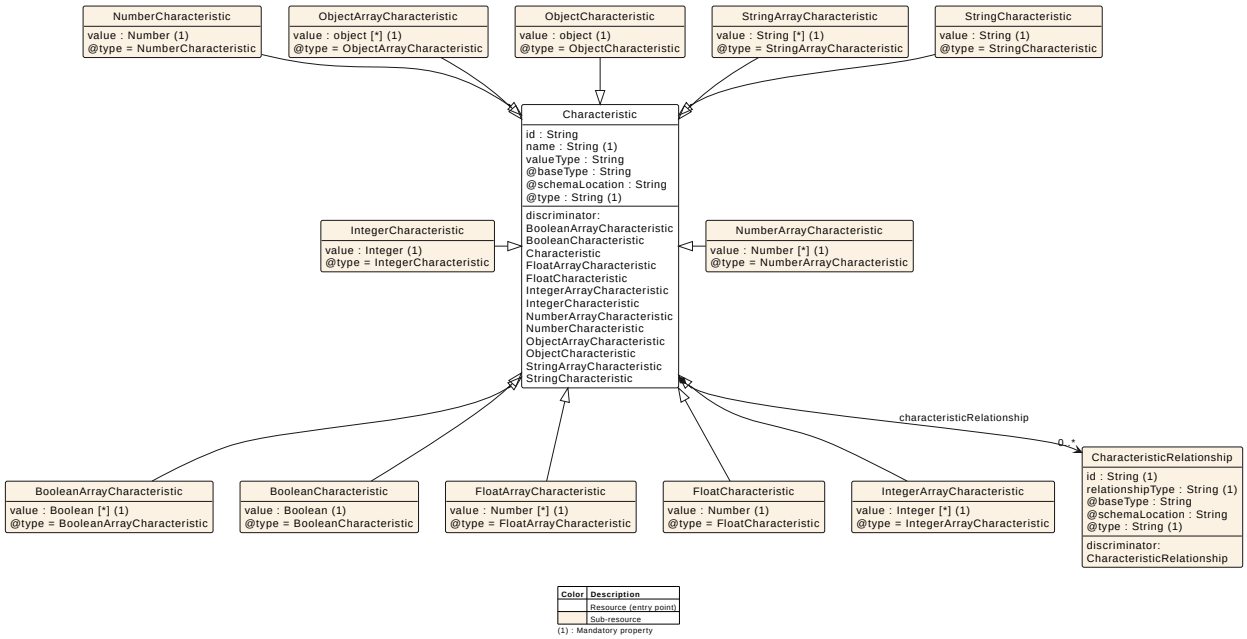


Figure 8 - Characteristic

© TM Forum 2026. All Rights Reserved.

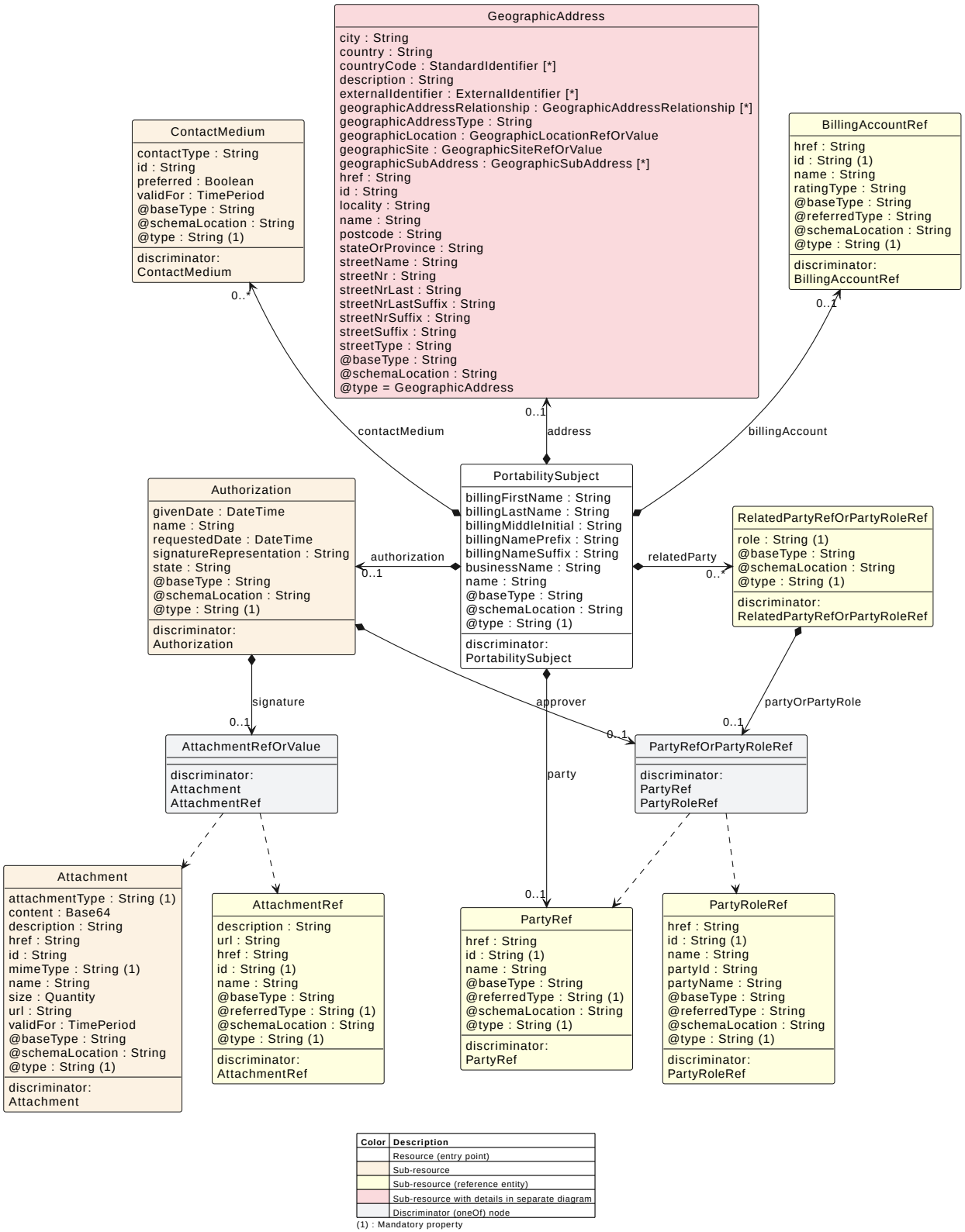


Figure 9 - PortabilitySubject

© TM Forum 2026. All Rights Reserved.

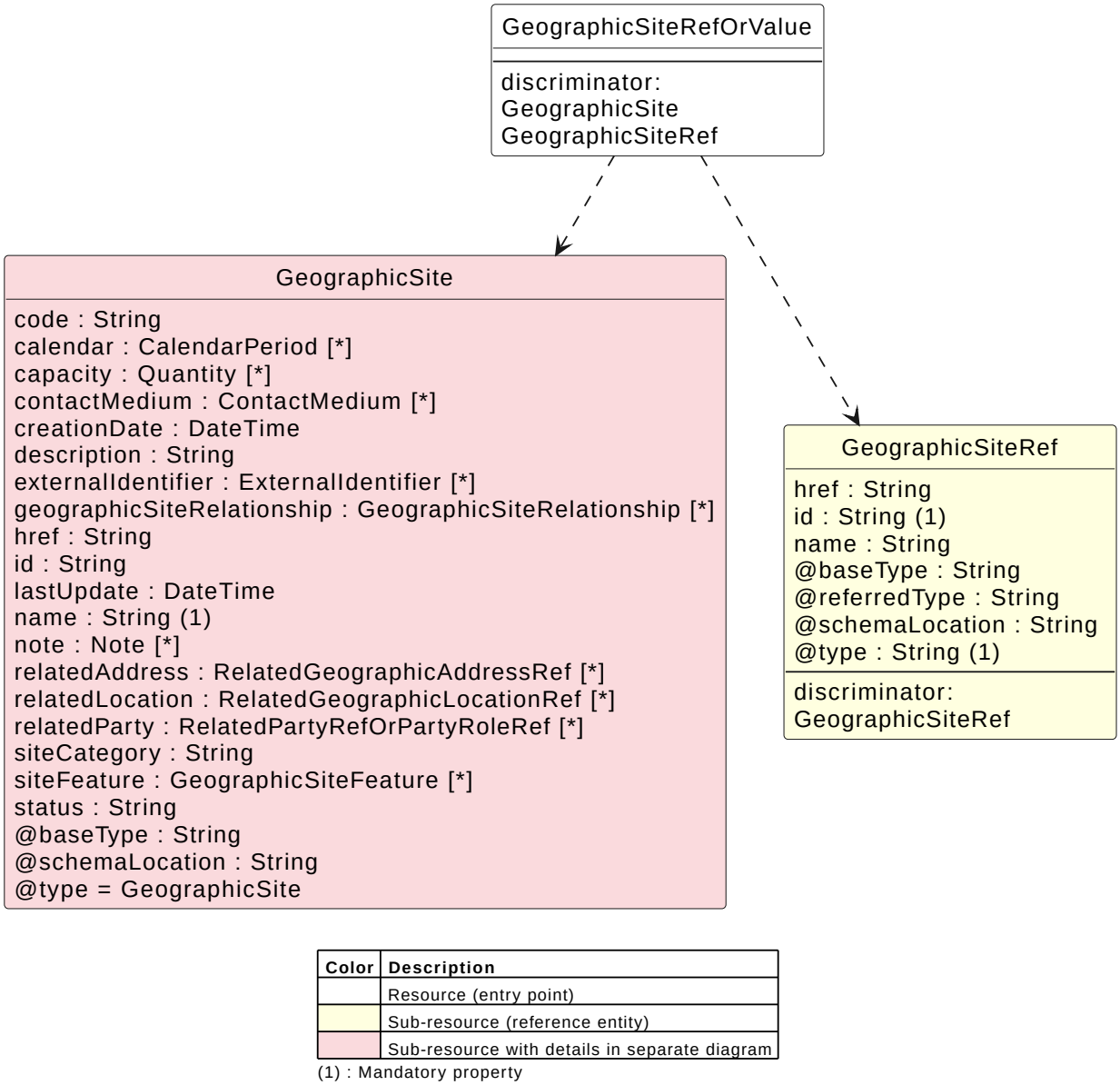


Figure 10 - GeographicSiteRefOrValue

© TM Forum 2026. All Rights Reserved.

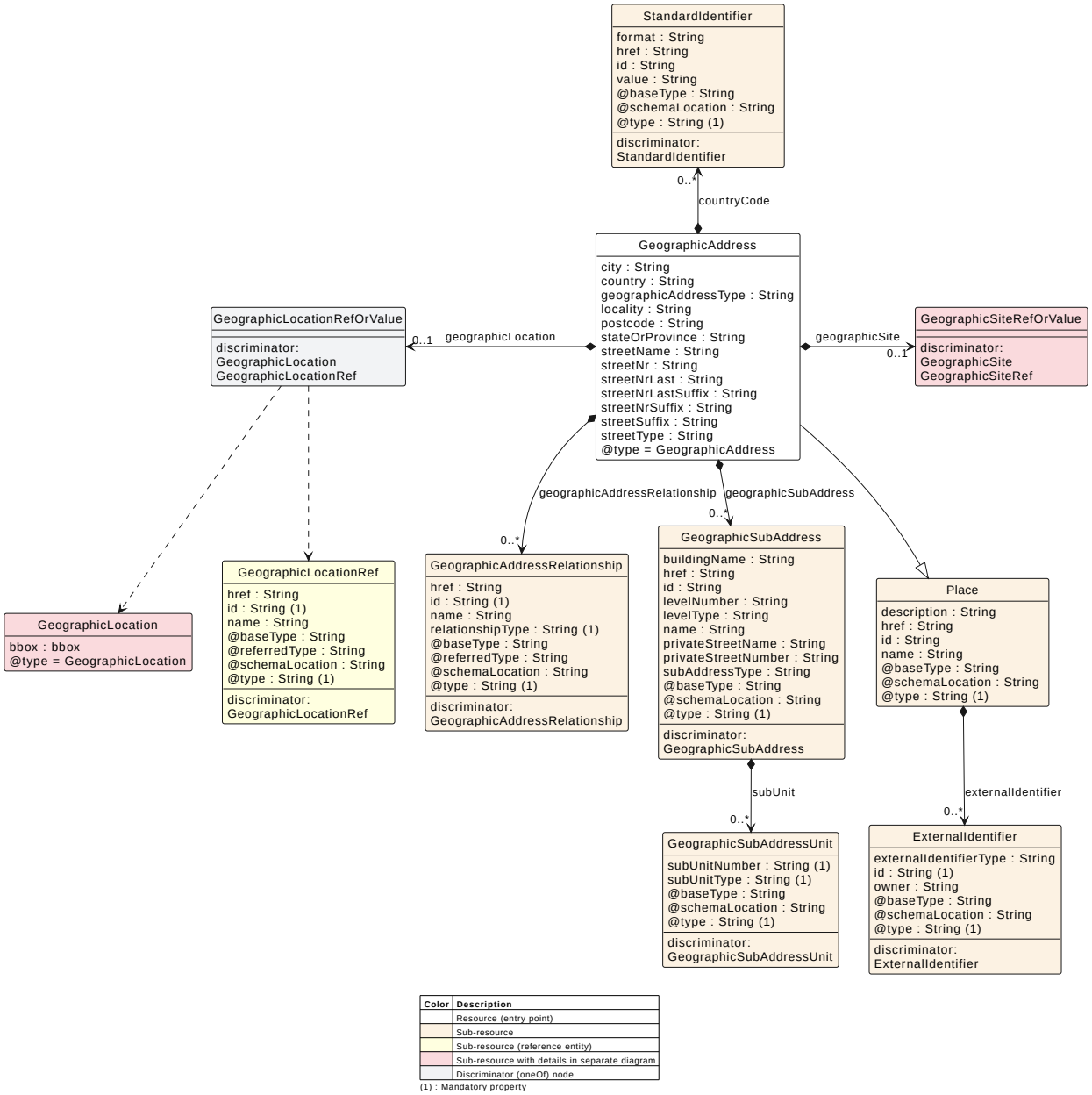


Figure 11 - GeographicAddress

© TM Forum 2026. All Rights Reserved.

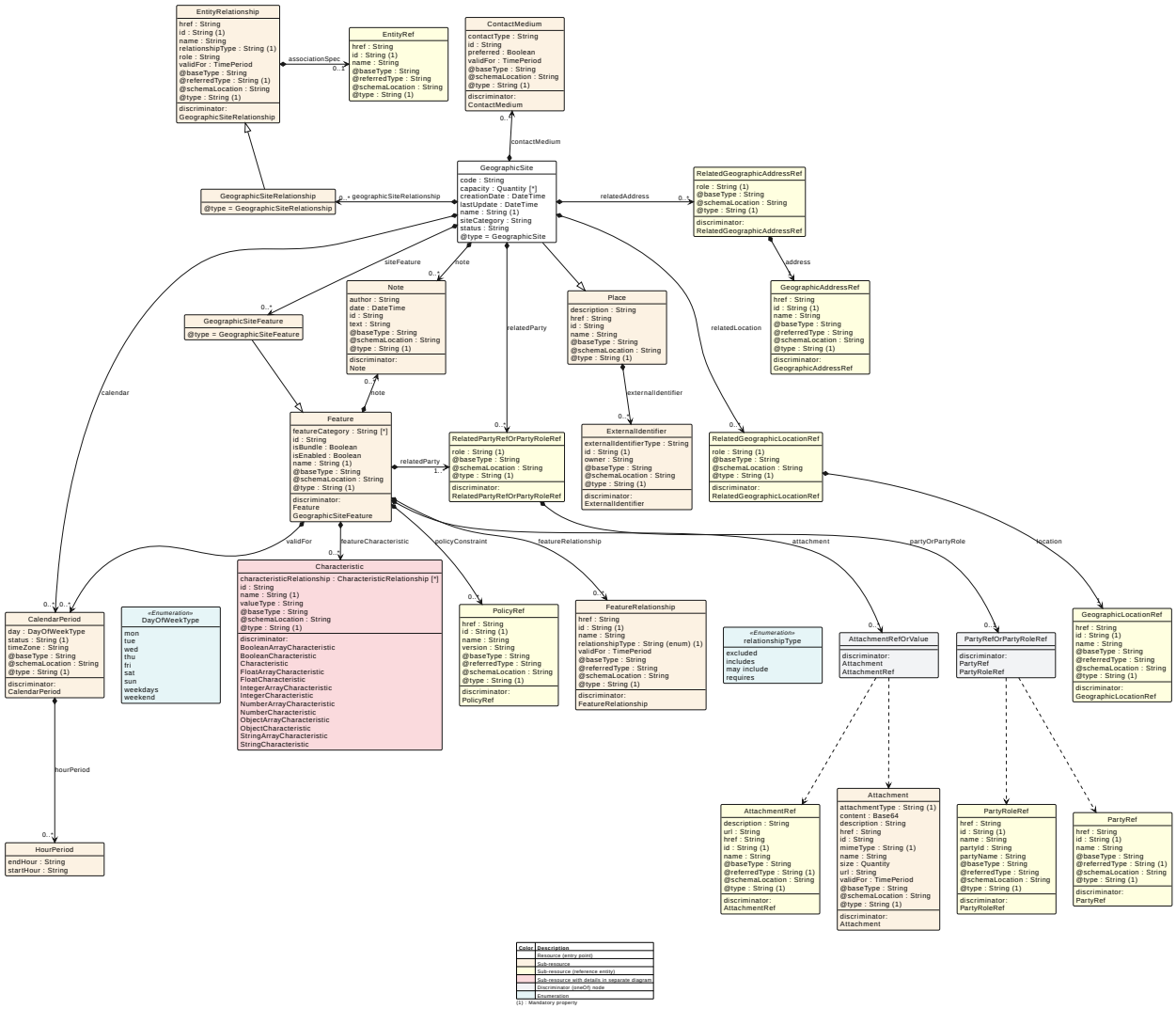


Figure 12 - GeographicSite

© TM Forum 2026. All Rights Reserved.

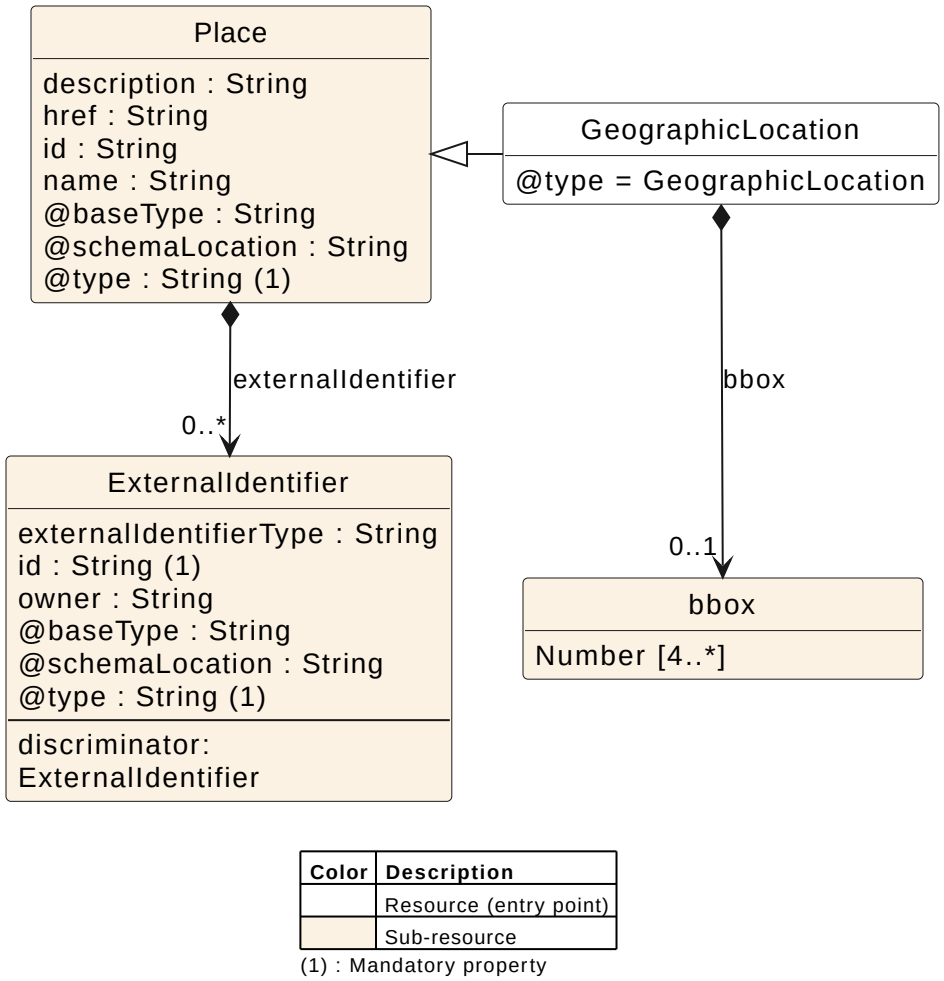


Figure 13 - GeographicLocation

© TM Forum 2026. All Rights Reserved.



Figure 14 - PhysicalResource

© TM Forum 2026. All Rights Reserved.

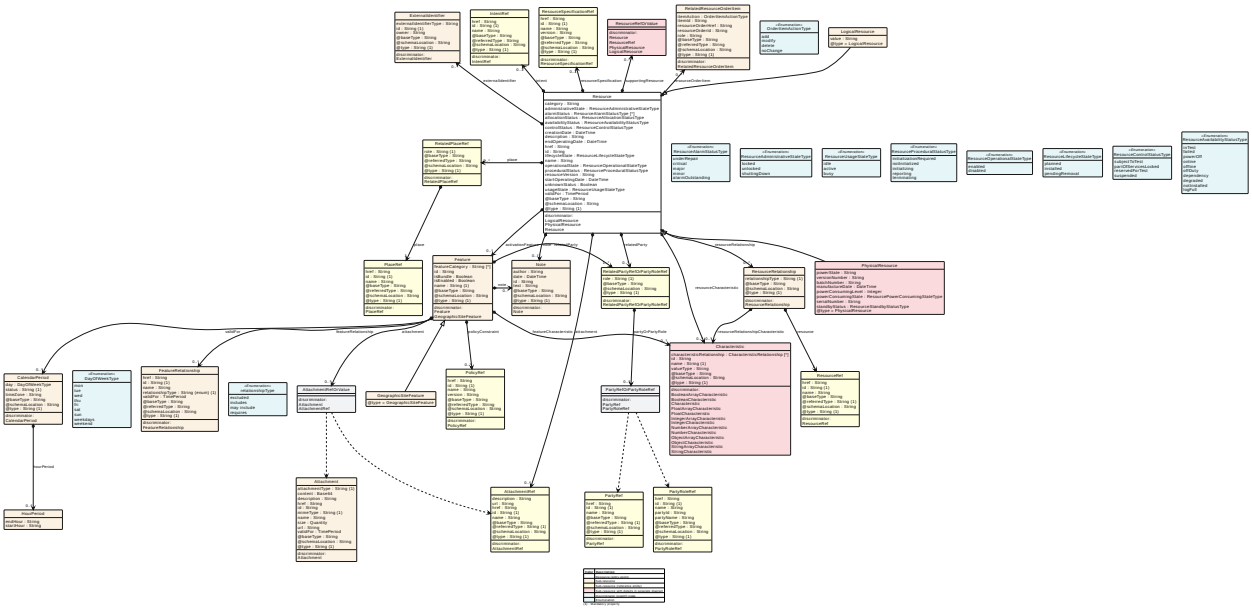


Figure 15 - Resource

© TM Forum 2026. All Rights Reserved.

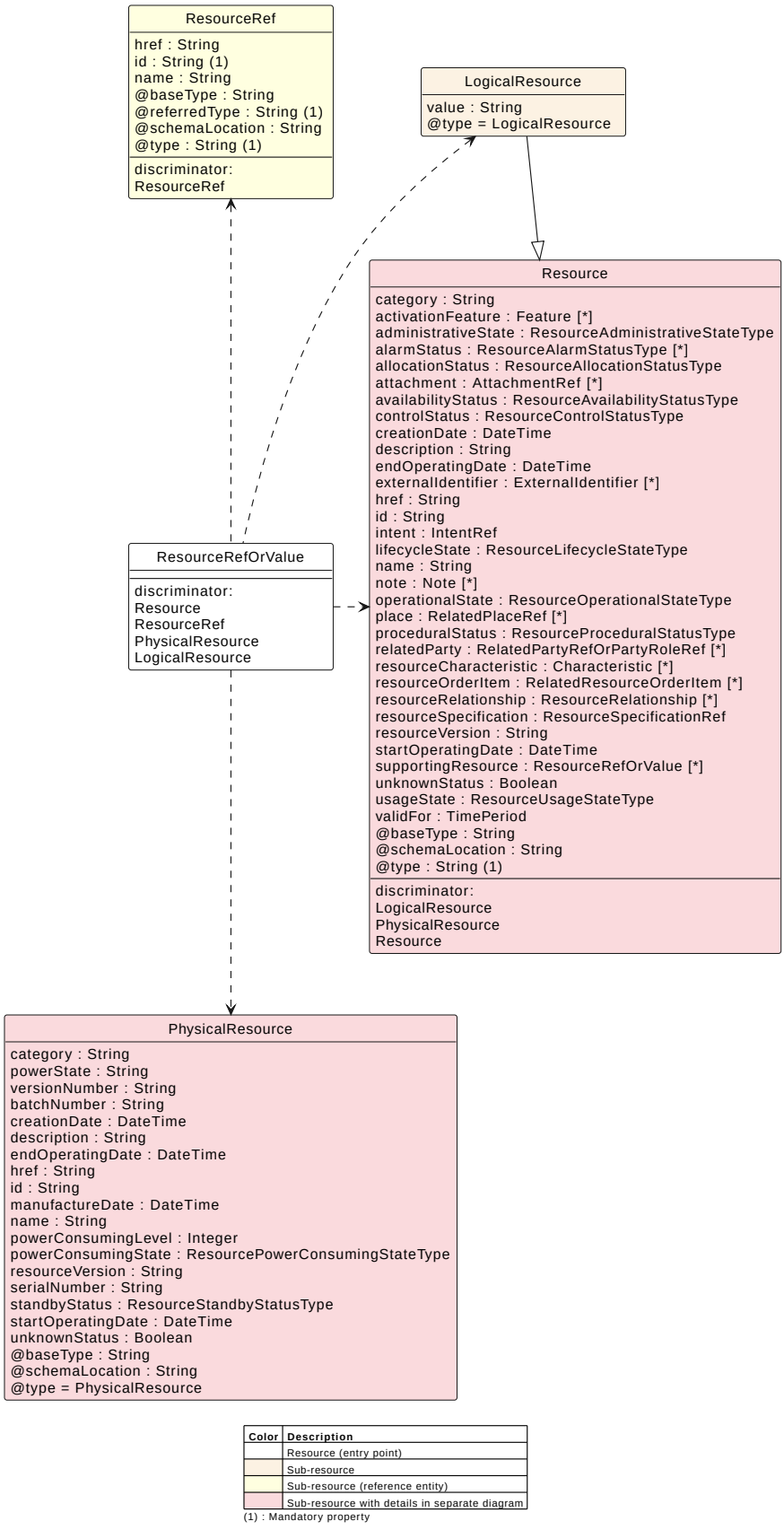


Figure 16 - ResourceRefOrValue

© TM Forum 2026. All Rights Reserved.

Field descriptions

PortabilityOrder fields

cancellationDate	A DateTime. Date when the order is cancelled - if cancelled, in RFC3339 (subset of ISO8601) format.
cancellationReason	A String. Reason why the order is cancelled - if cancelled.
category	A String. Used to categorize the order from a business perspective that can be useful for the order management system (e.g., 'Mobile Number Portability', 'Fixed Line Portability', 'Service Portability').
completionDate	A DateTime. Date when the order was actually completed, in RFC3339 (subset of ISO8601) format.
creationDate	A DateTime [1]. Date when the order was created, in RFC 3339 (subset of ISO8601) format.
description	A String. Free-text description of the Portability Order.
dispute	A Boolean. Indicates whether there is an active dispute associated with this portability order. A dispute may arise when the donor service provider challenges the portability request (e.g., due to outstanding balances, contract terms, or authorization issues).
donor	A ServiceProvider [1]. A Service Provider is a PartyRole played by a Party (typically an Organization) that provides telecommunications or digital services. In the context of portability, service providers act as either donors (losing the customer) or recipients (gaining the customer).
expectedCompletionDate	A DateTime. Date when the order is expected to be completed, in RFC3339 (subset of ISO8601) format.
externalId	An ExternalIdentifier [0..*]. Collection of external identifiers that reference this portability order in other systems (e.g., CRM systems, billing systems, or external portability databases).
groupRequestNumber	A String. Unique identifier for a group or batch of related portability requests that are processed together. This is used when multiple resources or services are being ported as part of a single coordinated request (e.g., porting multiple telephone numbers for a business customer).

© TM Forum 2026. All Rights Reserved.

href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. Date and time when the portability order was last updated, in RFC3339 (subset of ISO8601) format.
name	A String. A string used to give a name to the Portability Order.
note	A Note [0..*]. A list of notes containing additional human-readable information associated with the portability order.
orderItem	A PortabilityOrderItem [1..*]. A list of portability order items, each representing a discrete action to be performed as part of the portability order.
orderRelationship	An OrderRelationship [0..*]. A list of relationships to other orders.
portabilitySubject	A PortabilitySubject [1]. Contains information about the customer or account holder (the requestor) associated with the portability request. This entity aggregates authorization details, party information, and address to validate the portability request and ensure proper customer identification. Name and address information is denormalized to minimize inter-provider communication.
portRequestNumber	A String. Port Order Number (PON), also known as Port Request Number (PRN), is a unique identifier associated with this portability request. This reference number is used across service providers to track and coordinate the portability process. In some telecommunications markets, the PON is a standardized identifier used for regulatory compliance and inter-carrier coordination.
portRequestNumberVersion	A String. Version number of the port request, used to track revisions or amendments to the original portability request. This is incremented when changes are made to an existing port request.

© TM Forum 2026. All Rights Reserved.

priority	A PortabilityPriorityType [1]. Priority level for portability order processing. Regular priority follows standard processing timelines, while expedited priority requests faster processing. ENUMERATED with values: * regular * expedited
recipient	A ServiceProvider [1]. A Service Provider is a PartyRole played by a Party (typically an Organization) that provides telecommunications or digital services. In the context of portability, service providers act as either donors (losing the customer) or recipients (gaining the customer).
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of parties which are involved in this order and the role they are playing.
requestedCompletionDate	A DateTime. Requested delivery date from the requestor perspective, in RFC3339 (subset of ISO8601) format.
requestedStartDate	A DateTime. Order start date wished by the requestor, in RFC3339 (subset of ISO8601) format.
startDate	A DateTime. Date when the order was actually started, in RFC3339 (subset of ISO8601) format.
state	A PortabilityOrderStateType [1]. ENUMERATED with values: * acknowledged * pending * sent * delayed * rejected * held * confirmed * inProgress * cancelled * completed * failed * partial
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2026. All Rights Reserved.

ServiceProvider sub-resource fields

A Service Provider is a PartyRole played by a Party (typically an Organization) that provides telecommunications or digital services. In the context of portability, service providers act as either donors (losing the customer) or recipients (gaining the customer).

account	An AccountRef [0..*]. Collection of accounts.
agreement	An AgreementRef [0..*]. Collection of agreements.
characteristic	A Characteristic [0..*]. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium [0..*]. Collection of contact mediums.
creditProfile	A CreditProfile [0..*]. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef [1]. A Party reference.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.

© TM Forum 2026. All Rights Reserved.

paymentMethod	A PaymentMethodRef [0..*]. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole [0..*]. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

AccountRef sub-resource fields

Account reference. A account may be a party account or a financial account.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2026. All Rights Reserved.

AgreementRef sub-resource fields

Agreement reference. An agreement represents a contract or arrangement, either written or verbal and sometimes enforceable by law, such as a service level agreement or a customer price agreement. An agreement involves a number of other business entities, such as products, services, and resources and/or their specifications.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2026. All Rights Reserved.

	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

id	A String [1]. Unique identifier of the characteristic.
relationshipType	A String [1]. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1..*]. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.

© TM Forum 2026. All Rights Reserved.

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristic sub-resource fields

A characteristic which value is an array of Float(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Float [1..*]. A characteristic which value is an array of Float(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

A characteristic which value is a Float.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Float [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	An Integer [1..*]. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

© TM Forum 2026. All Rights Reserved.

A characteristic which value is a Integer.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	An Integer [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1..*]. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

A characteristic which value is a Number.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.

© TM Forum 2026. All Rights Reserved.

id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s). If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType [1..*]. A characteristic which value is an array of Object(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

A characteristic which value is an Object. If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

© TM Forum 2026. All Rights Reserved.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType_2 [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A String [1..*]. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

A characteristic which value is a String.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.

© TM Forum 2026. All Rights Reserved.

value	A String [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ContactMedium sub-resource fields

Indicates the contact medium that could be used to contact the party. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : EmailContact Medium, FaxContactMedium, PhoneContactMedium, GeographicAddressContactMedium, SocialMediaContactMedium...

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

CreditProfile sub-resource fields

Credit profile for the party (containing credit scoring, ...). By default only the current credit profile is retrieved. It can be used as a list to give the party credit profiles history, the first one in the list will be the current one.

creditProfileDate	A DateTime. The date the profile was established.
creditRiskRating	An Integer. This is an integer whose value is used to rate the risk.

© TM Forum 2026. All Rights Reserved.

creditScore	An Integer. A measure of a person or organizations creditworthiness calculated on the basis of a combination of factors such as their income and credit history.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

A Party reference.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1]. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRef

PartyRoleSpecificationRef sub-resource fields

Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.

© TM Forum 2026. All Rights Reserved.

@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PaymentMethodRef sub-resource fields

PaymentMethod reference. A payment method defines a specific mean of payment (e.g direct debit).

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyOrPartyRole sub-resource fields

A Party or PartyRole playing a specific role in context of the entity to which it is related, given by value or reference.

partyOrPartyRole	A PartyOrPartyRole. A concrete Party or a concrete PartyRole, by value or reference.
role	A String [1]. The role the Party or PartyRole plays in the specific context of the entity it is linked to, such as 'initiator', 'customer', 'salesAgent', 'user'; NOTE: If used for a PartyRole (by value or reference), the PartyRole describes the role the respective engagedParty plays in a more general context, like e.g. 'Customer', 'Manager'.
@baseType	A String. When sub-classing, this defines the super-class.

© TM Forum 2026. All Rights Reserved.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BusinessPartner sub-resource fields

When business partner is the BusinessPartner.

account	An AccountRef [0..*]. Collection of accounts.
agreement	An AgreementRef [0..*]. Collection of agreements.
characteristic	A Characteristic [0..*]. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium [0..*]. Collection of contact mediums.
creditProfile	A CreditProfile [0..*]. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef [1]. A Party reference.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.

© TM Forum 2026. All Rights Reserved.

partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef [0..*]. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole [0..*]. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Consumer sub-resource fields

A Consumer is a role that an individual, organization, or entity has if he/she/it uses services to satisfy his/her/its needs and wants.

account	An AccountRef [0..*]. Collection of accounts.
agreement	An AgreementRef [0..*]. Collection of agreements.

© TM Forum 2026. All Rights Reserved.

characteristic	A Characteristic [0..*]. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium [0..*]. Collection of contact mediums.
creditProfile	A CreditProfile [0..*]. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef [1]. A Party reference.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef [0..*]. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole [0..*]. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.

© TM Forum 2026. All Rights Reserved.

status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Individual sub-resource fields

Individual represents a single human being (a man, woman or child). The individual can be a customer, an employee or any other person that the organization needs to store information about.

aristocraticTitle	A String. E.g. Baron, Graf, Earl.
birthDate	A DateTime. Birth date.
contactMedium	A ContactMedium [0..*]. List of means for contacting the party, e.g. mobile phone, email address.
countryOfBirth	A String. Country where the individual was born.
creditRating	A PartyCreditProfile [0..*]. List of credit profiles and scores for the party, typically received from an external credit broker.
deathDate	A DateTime. Date of death.
disability	A Disability [0..*]. List of disabilities suffered by the individual.
externalReference	An ExternalIdentifier [0..*]. List of identifiers of the Party in an external system, for example when party information is imported from a commerce system.
familyName	A String. Contains the non-chosen or inherited name. Also known as last name in the Western context.
familyNamePrefix	A String. Family name prefix.

© TM Forum 2026. All Rights Reserved.

formattedName	A String. A fully formatted name in one string with all of its pieces in their proper place and all of the necessary punctuation. Useful for specific contexts (Chinese, Japanese, Korean).
gender	A String. Gender.
generation	A String. E.g.. Sr, Jr, III (the third).
givenName	A String. First name of the individual.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
individualIdentification	An IndividualIdentification [0..*]. List of official identifications issued to the individual, such as passport, driving licence, social security number.
languageAbility	A LanguageAbility [0..*]. List of national languages known by the individual.
legalName	A String. Legal name or birth name (name one has for official purposes).
location	A String. Temporary current location of the individual (may be used if the individual has approved its sharing).
maritalStatus	A String. Marital status (married, divorced, widow ...).
middleName	A String. Middles name or initial.
name	A String. Full name flatten (first, middle, and last names) - this is the name that is expected to be presented in reference data types such as PartyRef, RelatedParty, etc. that refer to Individual.
nationality	A String. Nationality.
otherName	An OtherNameIndividual [0..*]. List of other names by which this individual is known.

© TM Forum 2026. All Rights Reserved.

partyCharacteristic	A Characteristic [0..*]. List of additional characteristics that a Party can take on. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
placeOfBirth	A String. Reference to the place where the individual was born.
preferredGivenName	A String. Contains the chosen name by which the individual prefers to be addressed. Note: This name may be a name other than a given name, such as a nickname.
relatedParty	A RelatedPartyOrPartyRole [0..*]. List of parties and/or party roles related to this party.
skill	A Skill [0..*]. List of skills exhibited by the individual.
status	An IndividualStateType. Valid values for the lifecycle state of the individual. ENUMERATED with values: * initialized * validated * deceased
taxExemptionCertificate	A TaxExemptionCertificate [0..*]. List of tax exemptions granted to the party. For example, a war veteran might have partial exemption from state tax and a full exemption from federal tax.
title	A String. Useful for titles (aristocratic, social,...) Pr, Dr, Sir, ...
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PartyCreditProfile sub-resource fields

An individual might be evaluated for its worthiness and this evaluation might be based on a credit rating given by a credit agency.

creditAgencyName	A String. Name of the credit agency giving the score.
creditAgencyType	A String. Type of the credit agency giving the score.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
ratingReference	A String. Reference corresponding to the credit rating.
ratingScore	An Integer32. A measure of a party's creditworthiness calculated on the basis of a combination of factors such as their income and credit history.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Disability sub-resource fields

Lack or inadequate strength or ability.

disabilityCode	A String. Code of the disability.
disabilityName	A String. Name of the disability.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the

© TM Forum 2026. All Rights Reserved.

nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String [1]. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IndividualIdentification sub-resource fields

Represents our registration of information used as proof of identity by an individual (passport, national identity card, drivers license, social security number, birth certificate).

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
identificationId	A String. Identifier.
identificationType	A String. Identification type (passport, national identity card, drivers license, social security number, birth certificate).
issuingAuthority	A String. Authority which has issued the identifier, such as: social security, town hall.
issuingDate	A DateTime. Date at which the identifier was issued.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.

© TM Forum 2026. All Rights Reserved.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

Complements the description of an element (for instance a product) through video, pictures...

attachmentType	A String [1]. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, embedded inline and encoded as base64. Mutually exclusive with url.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String [1]. A technical characterization of the attachment content format using an IETF Media Type, for example application/pdf or image/png. Note: this property is named mimeType for historical reasons.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. The URL pointing to the attachment content held in an external system, such as a content management system or document repository. Mutually exclusive with content.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

~~description~~	DEPRECATED A String. A narrative text describing the content of the attachment. DEPRECATED: will be removed in v6.
href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
~~url~~	DEPRECATED A String. Link to the attachment media/content. DEPRECATED: will be removed in v6.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1]. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

LanguageAbility sub-resource fields

Ability of an individual to understand or converse in a language.

isFavouriteLanguage	A Boolean. A “true” value specifies whether the language is considered by the individual as his favourite one.
languageCode	A String. Language code (RFC 5646).
languageName	A String. Language name.
listeningProficiency	A String. Listening proficiency evaluated for this language.
readingProficiency	A String. Reading proficiency evaluated for this language.
speakingProficiency	A String. Speaking proficiency evaluated for this language.

© TM Forum 2026. All Rights Reserved.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
writingProficiency	A String. Writing proficiency evaluated for this language.

OtherNameIndividual sub-resource fields

Keeps track of other names, for example the old name of a woman before marriage or an artist name.

aristocraticTitle	A String. E.g. Baron, Graf, Earl, etc.
familyName	A String. Contains the non-chosen or inherited name. Also known as last name in the Western context.
familyNamePrefix	A String. Family name prefix.
formattedName	A String. . A fully formatted name in one string with all of its pieces in their proper place and all of the necessary punctuation. Useful for specific contexts (Chinese, Japanese, Korean, etc.).
fullName	A String. Full name flatten (first, middle, and last names).
generation	A String. E.g. Sr, Jr, etc.
givenName	A String. First name.
legalName	A String. Legal name or birth name (name one has for official purposes).
middleName	A String. Middle name or initial.
preferredGivenName	A String. Contains the chosen name by which the person prefers to be addressed. Note: This name may be a name other than a given name, such as a nickname.
title	A String. Use for titles (aristrocatic, social, ...): Pr, Dr, Sir,....
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

Skill sub-resource fields

Skills evaluated for an individual with a level and possibly with a limited validity when an obsolescence is defined (Ex: the first-aid certificate first level is limited to one year and an update training is required each year to keep the level).

© TM Forum 2026. All Rights Reserved.

comment	A String. A free text comment linked to the evaluation done.
evaluatedLevel	A String. Level of expertise in a skill evaluated for an individual.
skillCode	A String. Code of the skill.
skillName	A String. Name of the skill, such as Java language.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

TaxExemptionCertificate sub-resource fields

A tax exemption certificate represents a tax exemption granted to a party (individual or organization) by a tax jurisdiction which may be a city, state, country,... An exemption has a certificate identifier (received from the jurisdiction that levied the tax) and a validity period. An exemption is per tax types and determines for each type of tax what portion of the tax is exempted (partial by percentage or complete) via the tax definition.

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
certificateNumber	A String. Identifier of a document that shows proof of exemption from taxes for the taxing jurisdiction.
id	A String. Identifier of the tax exemption within list of the exemptions.
issuingJurisdiction	A String. Name of the jurisdiction that issued the exemption.
reason	A String. Reason for the tax exemption.
taxDefinition	A TaxDefinition [0..*]. A list of taxes that are covered by the exemption, e.g. City Tax, State Tax. The definition would include the exemption (e.g. for a rate exemption 0% would be a full exemption, 5% could be a partial exemption if the actual rate was 10%).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.

© TM Forum 2026. All Rights Reserved.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

TaxDefinition sub-resource fields

Reference of a tax definition. A tax is levied by an authorized tax jurisdiction. For example, there are many different types of tax (Federal Tax levied by the US Government, State Tax levied by the State of California, City Tax levied by the City of Los Angeles, etc.).

id	A String. Unique identifier of the tax.
jurisdictionLevel	A String. Level of the jurisdiction that levies the tax.
jurisdictionName	A String. Name of the jurisdiction that levies the tax.
name	A String. Tax name.
taxType	A String. Type of the tax.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Organization sub-resource fields

Organization represents a group of people identified by shared interests or purpose. Examples include business, department and enterprise. Because of the complex nature of many businesses, both organizations and organization units are represented by the same data.

contactMedium	A ContactMedium [0..*]. List of means for contacting the party, e.g. mobile phone, email address.
creditRating	A PartyCreditProfile [0..*]. List of credit profiles and scores for the party, typically received from an external credit broker.
existsDuring	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

© TM Forum 2026. All Rights Reserved.

externalReference	An ExternalIdentifier [0..*]. List of identifiers of the Party in an external system, for example when party information is imported from a commerce system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
isHeadOffice	A Boolean. If value is true, the organization is the head office.
isLegalEntity	A Boolean. If value is true, the organization is a legal entity known by a national referential.
name	A String. Organization name (department name for example).
nameType	A String. Type of the name : Co, Inc, Ltd, etc.
organizationChildRelationship	An OrganizationChildRelationship [0..*]. List of organizations that are contained within this organization. For example if this organization is the Legal Department, the child organizations might include Claims, Courts, Contracts.
organizationIdentification	An OrganizationIdentification [0..*]. List of official identifiers given to the organization, for example company number in the registry of companies.
organizationParentRelationship	An OrganizationParentRelationship. Parent references of an organization in a structure of organizations.
organizationType	A String. Type of Organization (company, department...).
otherName	An OtherNameOrganization [0..*]. List of additional names by which the organization is known.

© TM Forum 2026. All Rights Reserved.

partyCharacteristic	A Characteristic [0..*]. List of additional characteristics that a Party can take on. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
relatedParty	A RelatedPartyOrPartyRole [0..*]. List of parties and/or party roles related to this party.
status	An OrganizationStateType. Valid values for the lifecycle state of the organization. ENUMERATED with values: * initialized * validated * closed
taxExemptionCertificate	A TaxExemptionCertificate [0..*]. List of tax exemptions granted to the party. For example, a war veteran might have partial exemption from state tax and a full exemption from federal tax.
tradingName	A String. Name that the organization (unit) trades under.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

OrganizationChildRelationship sub-resource fields

Child references of an organization in a structure of organizations.

organization	An OrganizationRef. Reference to an Organization.

© TM Forum 2026. All Rights Reserved.

relationshipType	A String. Type of the relationship. Could be juridical, hierarchical, geographical, functional for example.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

OrganizationRef sub-resource fields

Reference to an Organization.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

OrganizationIdentification sub-resource fields

Represents our registration of information used as proof of identity by an organization.

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
identificationId	A String. Identifier.
identificationType	A String. Type of identification information used to identify the company in a country or internationally.
issuingAuthority	A String. Authority which has issued the identifier (chamber of commerce...).
issuingDate	A DateTime. Date at which the identifier was issued.

© TM Forum 2026. All Rights Reserved.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

OrganizationParentRelationship sub-resource fields

Parent references of an organization in a structure of organizations.

organization	An OrganizationRef. Reference to an Organization.
relationshipType	A String. Type of the relationship. Could be juridical, hierarchical, geographical, functional for example.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

OtherNameOrganization sub-resource fields

Keeps track of other names, for example the old name of an organization.

name	A String. Organization name (department name for example).
nameType	A String. Co. , Inc. , Ltd. , Pty Ltd. , Plc; , Gmbh.
tradingName	A String. The name that the organization trades under.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2026. All Rights Reserved.

PartyRole sub-resource fields

The part played by a party in a given context.

account	An AccountRef [0..*]. Collection of accounts.
agreement	An AgreementRef [0..*]. Collection of agreements.
characteristic	A Characteristic [0..*]. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium [0..*]. Collection of contact mediums.
creditProfile	A CreditProfile [0..*]. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef [1]. A Party reference.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef [0..*]. Collection of payment methods.

© TM Forum 2026. All Rights Reserved.

relatedParty	A RelatedPartyOrPartyRole [0..*]. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	PartyRole can be instantiated as * BusinessPartner * Consumer * Producer * ServiceProvider * Supplier
value	This property is present in subclasses

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.

© TM Forum 2026. All Rights Reserved.

partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRoleRef

Producer sub-resource fields

A Producer is a role that an individual, organization, or entity has if he/she/it produces resources or services that Consumers may use to satisfy their needs and wants.

account	An AccountRef [0..*]. Collection of accounts.
agreement	An AgreementRef [0..*]. Collection of agreements.
characteristic	A Characteristic [0..*]. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium [0..*]. Collection of contact mediums.
creditProfile	A CreditProfile [0..*]. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef [1]. A Party reference.

© TM Forum 2026. All Rights Reserved.

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef [0..*]. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole [0..*]. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.
status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Supplier sub-resource fields

When business partner is the Supplier.

account	An AccountRef [0..*]. Collection of accounts.
agreement	An AgreementRef [0..*]. Collection of agreements.

© TM Forum 2026. All Rights Reserved.

characteristic	A Characteristic [0..*]. Describes the characteristic of a party role. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
contactMedium	A ContactMedium [0..*]. Collection of contact mediums.
creditProfile	A CreditProfile [0..*]. Collection of credit profiles.
description	A String. A description of the PartyRole.
engagedParty	A PartyRef [1]. A Party reference.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity.
partyRoleSpecification	A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.
paymentMethod	A PaymentMethodRef [0..*]. Collection of payment methods.
relatedParty	A RelatedPartyOrPartyRole [0..*]. Collection of related parties.
role	A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification.

© TM Forum 2026. All Rights Reserved.

status	A String. Used to track the lifecycle status of the party role.
statusReason	A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

Extra information about a given entity.

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PortabilityOrderItem sub-resource fields

An identified part of the portability order. Each order item represents a discrete resource to be ported. Portability orders contain a flat list of order items - there is no hierarchy of items.

action	A PortabilityActionType. ENUMERATED with values: * disconnect * port * intraport

© TM Forum 2026. All Rights Reserved.

endUserName	A String. Name of the actual end user of the resource. This may differ from the account holder (portabilitySubject) in business scenarios where resources are assigned to employees, departments, or sub-accounts. Used for validation and authorization purposes.
id	A String [1]. Identifier of the order item (generally a sequence number: 1, 2, 3, etc.).
message	An ErrorMessage [0..*]. Processing messages, error codes, and status updates specific to this order item. These messages track the technical progress of the portability action and facilitate troubleshooting.
note	A Note [0..*]. Additional notes or comments specific to this order item.
portabilityActivationMode	A PortabilityActivationModeType. Specifies how the resource activation occurs at the recipient provider. Automatic activation happens at the scheduled time without manual intervention, while manual activation requires explicit action for coordinated cutover scenarios. ENUMERATED with values: * automatic * manual
portabilityEffectiveDate	A DateTime. The exact date and time when the portability takes effect - when the resource becomes active at the recipient provider and billing ceases at the donor provider. This date is critical for regulatory compliance and billing accuracy.
resource	A Resource. Resource is an abstract entity that describes the common set of attributes shared by all concrete resources (e.g. TPE, EQUIPMENT) in the inventory. Resource can be instantiated as * LogicalResource * PhysicalResource

© TM Forum 2026. All Rights Reserved.

state	A PortabilityOrderItemStateType. ENUMERATED with values: * acknowledged * rejected * pending * confirmed * executed * deleted * cancelled
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ErrorMessage sub-resource fields

Represents an Error.

code	A String. Error code.
message	A String. More details and corrective actions related to the error.
reason	A String. Explanation of the reason for the error.
referenceError	A String. URI of documentation describing the error.
status	A String. Error code extension like sys-ABC-2001.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Resource sub-resource fields

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources (e.g. TPE, EQUIPMENT) in the inventory.

activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.

© TM Forum 2026. All Rights Reserved.

administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

© TM Forum 2026. All Rights Reserved.

alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

© TM Forum 2026. All Rights Reserved.

availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

© TM Forum 2026. All Rights Reserved.

~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.

© TM Forum 2026. All Rights Reserved.

externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled

© TM Forum 2026. All Rights Reserved.

place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating

© TM Forum 2026. All Rights Reserved.

relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.

© TM Forum 2026. All Rights Reserved.

resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.

© TM Forum 2026. All Rights Reserved.

usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	Resource can be instantiated as * LogicalResource * PhysicalResource
value	This property is present in subclasses

Feature sub-resource fields

Configuration feature.

attachment	An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation.
featureCategory	A String [0..*]. Collection of feature categories.

© TM Forum 2026. All Rights Reserved.

featureCharacteristic	A Characteristic [0..*]. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship [0..*]. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String [1]. The name for the feature.
note	A Note [0..*]. Notes relevant to this Feature.
policyConstraint	A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature.
validFor	A CalendarPeriod [0..*]. Validity period of this Feature.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	Feature can be instantiated as * GeographicSiteFeature
value	This property is present in subclasses

© TM Forum 2026. All Rights Reserved.

FeatureRelationship sub-resource fields

Configuration feature.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. This is the name of the target feature.
relationshipType	A String [1]. This is the type of the feature relationship. ENUMERATED with values: * excluded * includes * may include * requires
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

Reference to managed Policy object.

href	A String. Hyperlink reference.
id	A String [1]. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.

partyOrPartyRole	A PartyRefOrPartyRoleRef. A reference to a Party or a PartyRole.
role	A String [1]. The role the referred Party or PartyRole plays in the specific context of the entity it is linked to, such as 'initiator', 'customer', 'salesAgent', 'user'; NOTE: If used to refer to a PartyRole, the PartyRole describes the role the respective engagedParty plays in a more general context, like e.g. 'Customer', 'Manager'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

day	A DayOfWeekType. Possible values for the days of the week. ENUMERATED with values: * mon * tue * wed * thu * fri * sat * sun * weekdays * weekend

© TM Forum 2026. All Rights Reserved.

hourPeriod	A HourPeriod [0..*]. Start and end time range in 24-hour HH:mm format.
status	A String [1]. Indication of the availability of the calendar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information in ISO 8601 timezone: either 'Z' for UTC or an offset like +02:00 or -05:30.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

HourPeriod sub-resource fields

Hour interval.

endHour	A String. The end of the hour period, expressed in 24-hour HH:mm format (e.g., 12:00 for noon, 23:59 for one minute before midnight).
startHour	A String. The beginning of the hour period, expressed in 24-hour HH:mm format (e.g., 09:00 for 9 AM, 17:30 for 5:30 PM).

GeographicSiteFeature sub-resource fields

Geographic Site Feature captures various site information, ranging from survey data to safety guidelines and hazard information.

attachment	An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation.
featureCategory	A String [0..*]. Collection of feature categories.

© TM Forum 2026. All Rights Reserved.

featureCharacteristic	A Characteristic [0..*]. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship [0..*]. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String [1]. The name for the feature.
note	A Note [0..*]. Notes relevant to this Feature.
policyConstraint	A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature.
validFor	A CalendarPeriod [0..*]. Validity period of this Feature.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntentRef sub-resource fields

Intent reference, for when Intent is used by other entities.

© TM Forum 2026. All Rights Reserved.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRef sub-resource fields

Use role to distinguish the context of the place being referenced (such as `delivery-entrance`). Use `@referredType` to distinguish the specific Place sub-type (such as a `GeographicAddress`). Use `@type` to extend (eg: as RefOrValue).

place	A PlaceRef [1]. A reference to a physical Place.
role	A String [1]. The context in which this Place is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The kind of Place that is being referenced.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

A reference to a physical Place.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The specific sub-type of Place being referred to (such as a `GeographicAddress`).

© TM Forum 2026. All Rights Reserved.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedResourceOrderItem sub-resource fields

RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.

itemAction	An OrderItemActionType. Action to be performed on the entity managed by the item. ENUMERATED with values: * add * modify * delete * noChange
itemId	A String. Identifier of the order item where the resource was managed.
resourceOrderHref	A String. Reference of the related entity.
resourceOrderId	A String. Unique identifier of a related entity.
role	A String. Role of the resource order item for this resource.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceRelationship sub-resource fields

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

© TM Forum 2026. All Rights Reserved.

relationshipType	A String [1]. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [relies On] if the resource needs another already owned resource to rely on (eg: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A ResourceRef [1]. Resource reference, for when Resource is used by other entities.
resourceRelationshipCharacteristic	A Characteristic [0..*]. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic... Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2026. All Rights Reserved.

ResourceRef sub-resource fields

Resource reference, for when Resource is used by other entities.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PhysicalResource sub-resource fields

Physical resource is a type of resource that describes the common set of attributes shared by all concrete physical resources (e.g. EQUIPMENT) in the inventory.

activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

© TM Forum 2026. All Rights Reserved.

alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

© TM Forum 2026. All Rights Reserved.

availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

© TM Forum 2026. All Rights Reserved.

batchNumber	A String. A unique identifier assigned to a specific group of physical resources produced or received under the same conditions for tracking and traceability in the supply chain.
~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.

© TM Forum 2026. All Rights Reserved.

endOperatingDate	A DateTime. The point in time till the resource is operating.
externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
manufactureDate	A DateTime. This is a string attribute that defines the date of manufacture of this item in the ISO 8601 date time standard format. This is an optional attribute.
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.

© TM Forum 2026. All Rights Reserved.

operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled
place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
powerConsumingLevel	An Integer. The power consuming level attribute (defined by ITU-T M.3701) indicates the current percentage of power consuming of the resource represented by the managed object, where 100 percent represents the maximum power consumption of the device. It is single valued and read-only by management functions.

© TM Forum 2026. All Rights Reserved.

powerConsumingState	A ResourcePowerConsumingStateType. Value enumeration for the 'resourcePowerConsumingState' of a Resource; the power consuming state attribute (defined by ITU-T M.3701) indicates the power consuming policy of the resource represented by the managed object, it is single valued and read-write by management functions.; defined values: 'fullPower': The resource runs in full power and will not consider power saving. If a large device contains parts that also support this attribute, when the container object set to full power state, all its parts shall turn into full power state; 'powerSaving': The resource is (or may be) partially running to save power; depending on how much the payload is lower than the maximum capacity of the resource, the degree of saving will be automatically decided by the resource. If power saving is not supported by the resource, attempting to change to this state will have the same effect as full power state. If a large device contains parts that also support this attribute, when the container object is set to power saving state, its parts may still remain in full power state. But if one of the parts turns into power saving state, the container object shall turn into power saving state; 'sleeping': The resource is not running but keeps a snapshot of its memory before sleeping, and can resume running mode quickly. In this mode, the power is only consumed to meet the need of maintaining the memory. If it is not supported by the resource, attempting to change to this state will result in a failure; 'shutdown': The resource has been powered off and does not consume any more power; if it is not supported by the resource, attempting to change to this state will result in a failure. ENUMERATED with values: * fullPower * powerSaving * sleeping * shutdown

© TM Forum 2026. All Rights Reserved.

~~powerState~~	DEPRECATED A String. This defines the current power status of the hardware item. Values include: 0: Unknown 1: Not Applicable 2: No Power Applied 3: Full Power Applied 4: Power Save - Normal 5: Power Save - Degraded 6: Power Save - Standby 7: Power Save - Critical 8: Power Save - Low Power Mode 9: Power Save - Unknown 10: Power Cycle 11: Power Warning 12: Power Off; NOTE: This attribute is replaced by the 'powerConsuming State' attribute, to align with the official name as defined by ITU-T.

© TM Forum 2026. All Rights Reserved.

proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).

© TM Forum 2026. All Rights Reserved.

resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

© TM Forum 2026. All Rights Reserved.

resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
serialNumber	A String. This is a string that represents a manufacturer-allocated number used to identify different instances of the same hardware item. The ModelNumber and Part Number attributes are used to identify different types of hardware items.
standbyStatus	A ResourceStandbyStatusType. Value enumeration for the 'resourceStandbyStatus' of a Resource; the standby status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the working mode for standby objects, i.e. whether it is providing service or standby as a back-up resource; it is single valued and read-only by management functions; defined values: 'hotStandby': The resource is not providing service, but is operating in synchronism with another resource that is to be backed-up (e.g., a computer shadowing another computer). A resource with a hot standby status will be immediately able to take over the role of the resource to be backed up, without the need for initialization activity, and will contain the same information as the resource to be backed up; 'coldStandby': The resource is to back up another resource, but is not synchronized with that resource. A resource with a cold standby status will not be immediately able to take over the role of a resource to be backed up, and will require some initialization activity; 'providingService': The back up resource is providing service and is backing up another resource. ENUMERATED with values: * hotStandby * coldStandby * providingService
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).

© TM Forum 2026. All Rights Reserved.

unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
~~versionNumber~~	DEPRECATED A String. This is a string that identifies the version of this physical resource. This is an optional attribute. NOTE that this attribute is deprecated as it overlaps with the Resource.resourceVersion attribute.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

LogicalResource sub-resource fields

Logic resource is a type of resource that describes the common set of attributes shared by all concrete logical resources (e.g. TPE, MSISDN, IP Addresses) in the inventory.

© TM Forum 2026. All Rights Reserved.

activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

© TM Forum 2026. All Rights Reserved.

alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

© TM Forum 2026. All Rights Reserved.

availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

© TM Forum 2026. All Rights Reserved.

~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.

© TM Forum 2026. All Rights Reserved.

externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled

© TM Forum 2026. All Rights Reserved.

place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating

© TM Forum 2026. All Rights Reserved.

relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.

© TM Forum 2026. All Rights Reserved.

resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.

© TM Forum 2026. All Rights Reserved.

usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

OrderRelationship sub-resource fields

Used to describe relationship between order.

href	A String. The URI of the referred entity.
id	A String [1]. Id of the related order.
name	A String. Name of the referred entity.
relationshipType	A String [1]. Relationship type as correlates, triggers, etc...
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

© TM Forum 2026. All Rights Reserved.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PortabilitySubject sub-resource fields

Contains information about the customer or account holder (the requestor) associated with the portability request. This entity aggregates authorization details, party information, and address to validate the portability request and ensure proper customer identification. Name and address information is denormalized to minimize inter-provider communication.

address	A GeographicAddress. Structured textual way of describing how to find a Property in an urban area (country properties are often defined differently). Note : Address corresponds to SID UrbanPropertyAddress.
authorization	An Authorization. If special discount or special product offering price or specific condition need an approval for ISP sale representative it is described here.
billingAccount	A BillingAccountRef. BillingAccount reference. A BillingAccount is a detailed description of a bill structure.
billingFirstName	A String. First name of the billing account holder.
billingLastName	A String. Last name of the billing account holder.
billingMiddleInitial	A String. Middle initial of the billing account holder.
billingNamePrefix	A String. Name prefix (e.g., Mr., Mrs., Dr.) for the billing account holder.
billingNameSuffix	A String. Name suffix (e.g., Jr., Sr., III) for the billing account holder.
businessName	A String. Business or organization name if the portability subject is a business customer.
contactMedium	A ContactMedium [0..*]. Contact information for the customer or account holder.
name	A String. Full name of the customer or account holder associated with the portability request. This is denormalized from the Party for convenience and to avoid chatty interfaces between service providers.
party	A PartyRef. A Party reference.

© TM Forum 2026. All Rights Reserved.

relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. Additional parties involved in the portability request (e.g., authorized representatives, legal guardians).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicAddress sub-resource fields

Structured textual way of describing how to find a Property in an urban area (country properties are often defined differently). Note : Address corresponds to SID UrbanProperty Address.

city	A String. City that the address is in.
country	A String. Country that the address is in.
countryCode	A StandardIdentifier [0..*]. Country codes are short alphabetic or numeric geographical codes (geocodes) developed to represent countries and dependent areas, for use in data processing and communications. Several different systems have been developed to do this. The term country code frequently refers to ISO 3166-1 alpha-2 or international dialing codes, the E.164 country calling codes.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geographicAddressRelationship	A GeographicAddressRelationship [0..*]. A list of geographic addresses relationships. Represents a relationship between geographic addresses.
geographicAddressType	A String. Classification of the address, e.g., residential, industrial.

© TM Forum 2026. All Rights Reserved.

geographicLocation	A GeographicLocationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicLocation entity and not the GeographicLocationRefOrValue class itself.
geographicSite	A GeographicSiteRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicSite entity and not the GeographicSiteRefOrValue class itself.
geographicSubAddress	A GeographicSubAddress [0..*]. Collection of sub addresses.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
locality	A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature. [ANZLIC-STREET], or a suburb, a bounded locality within a city, town or shire principally of urban character [ANZLICSTREET].
name	A String. A user-friendly name for the Place.
postcode	A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode).
stateOrProvince	A String. The State or Province that the address is in.
streetName	A String. Name of the street or other street type.
streetNr	A String. Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.
streetNrLast	A String. Last number in a range of street numbers allocated to a property.
streetNrLastSuffix	A String. Last street number suffix for a ranged address.
streetNrSuffix	A String. The first street number suffix.
streetSuffix	A String. A modifier denoting a relative direction.
streetType	A String. Alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf.
@baseType	A String. When sub-classing, this defines the super-class.

© TM Forum 2026. All Rights Reserved.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StandardIdentifier sub-resource fields

The corresponding identification of the resource in different standard, regulatory definitions. The standard specification identifier (e.g., ISO 3166-1 Alpha-2) and the corresponding value (e.g., BE) relevant to a particular resource. It is anticipated that multiple standards can provide definitions for a single entity, e.g., a country identifier can be specified in various standards (e.g., "ISO 3166-1 Alpha 2", "ISO 3166-1 Alpha 3", "ISO 3166-1 Numeric").

format	A String. Standard/Regulatory definition identifier. e.g., ISO 3166-1.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
value	A String. The value of the resource in the corresponding standard.e.g., a country code value.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressRelationship sub-resource fields

The GeographicAddressRelationship schema represents a relationship between geographic addresses. It defines the structure for storing information about how two geographic addresses are related to each other within a system.

href	A String. The URI of the referred entity.
id	A String [1]. Unique identifier of the related geographic address. This identifier serves as a reference to uniquely identify the associated geographic address within the system.

© TM Forum 2026. All Rights Reserved.

name	A String. Name of the geographic address. This attribute provides a human-readable name or label for the related geographic address, which can help users identify and differentiate between different addresses.
relationshipType	A String [1]. Type of relationship between the geographic addresses. This attribute specifies the nature of the relationship between the related addresses. It provides context for understanding how the addresses are related to each other.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

A GeographicLocation is a pure-virtual super-class to the GeoJSON-aligned geometries of Point (addresses and locations), MultiPoint, LineString (streets, highways and boundaries), MultiLineString and Polygon (countries, provinces, tracts of land). Use the @type attribute to specify which of these is being specified by the geometry attribute.

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicLocationRef sub-resource fields

Reference to a Geographic Location.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicSite sub-resource fields

A GeographicSite is a managed place, such as a retail store, warehouse or data centre. It will have related responsible parties, a calendar of access times and be described by addresses and/or locations.

calendar	A CalendarPeriod [0..*]. Periods of (for example) access, opening, alarm, unavailability of this site.
capacity	A Quantity [0..*]. Capacities for the site indicating maximum operational thresholds (for example: people, floorspace, racks).
~~code~~	DEPRECATED A String. Deprecated: Use id, externalIdentifier and/or site Category or name.
contactMedium	A ContactMedium [0..*]. Contact information for people with responsibility for this site.
creationDate	A DateTime. Date and time when the GeographicSite was created.
description	A String. Additional information of a user-friendly description of the Place.

© TM Forum 2026. All Rights Reserved.

externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geographicSiteRelationship	A GeographicSiteRelationship [0..*]. This site might have relationships to other sites.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. Date and time when the GeographicSite was last updated.
name	A String [1]. A user-friendly name for the Place.
note	A Note [0..*]. Unstructured notes made about this site.
relatedAddress	A RelatedGeographicAddressRef [0..*]. Collection of related address references (such as delivery-address, office-address).
relatedLocation	A RelatedGeographicLocationRef [0..*]. Collection of related location references (such as retail-complex, service-access).
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related roles and parties references (such as site manager, site owner, site operator).
siteCategory	A String. Site classification/category.
siteFeature	A GeographicSiteFeature [0..*]. Collection of site features.
status	A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRelationship sub-resource fields

Details of a geographic site relationship.

© TM Forum 2026. All Rights Reserved.

associationSpec	An EntityRef. Entity reference schema to be use for all entityRef class.
href	A String.
id	A String [1].
name	A String.
relationshipType	A String [1]. Type of relationship between GeographicSites, such as adjacentTo, connectedBy.
role	A String. The association role for this entity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1].
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1].

EntityRef sub-resource fields

Entity reference schema to be use for all entityRef class.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedGeographicAddressRef sub-resource fields

A Postal Address (by reference) with its associated role.

© TM Forum 2026. All Rights Reserved.

address	A GeographicAddressRef [1]. Reference to a Geographic Address.
role	A String [1]. The context in which this Address is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressRef sub-resource fields

Reference to a Geographic Address.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedGeographicLocationRef sub-resource fields

A Geographic Location (by reference) with its associated role.

location	A GeographicLocationRef [1]. Reference to a Geographic Location.
role	A String [1]. The context in which this location is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRef sub-resource fields

Entity reference schema to be use for all entityRef class.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddress sub-resource fields

Representation of a GeographicSubAddress It is used for addressing within a property in an urban area (country properties are often defined differently). It may refer to a building, a building cluster, or a floor of a multistory building.

buildingName	A String. Allows for buildings that have well-known names.
href	A String. Link to the subAddress.
id	A String. Unique Identifier of the subAddress.
levelNumber	A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2.
levelType	A String. Describes level types within a building.
name	A String. Name of the subAddress to identify it with a meaningful identification.
privateStreetName	A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office.
privateStreetNumber	A String. Private streets numbers internal to a private street.

© TM Forum 2026. All Rights Reserved.

subAddressType	A String. Type of subAddress : it can be a subunit or a private street.
subUnit	A GeographicSubAddressUnit [0..*]. Representation of a SubUnit. It is used for describing subunit within a sub Address e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddressUnit sub-resource fields

Representation of a SubUnit. It is used for describing subunit within a subAddress e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.

subUnitNumber	A String [1]. The discriminator used for the subunit, often just a simple number but may also be a range.
subUnitType	A String [1]. The type of subunit e.g.BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF, RACK.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Place sub-resource fields

Place is the abstract base-class of GeographicAddress, GeographicSite and Geographic Location. There is no Place API.

description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
href	A String. Hyperlink reference.

© TM Forum 2026. All Rights Reserved.

id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	Place can be instantiated as * GeographicAddress * GeographicLocation * GeographicSite
value	This property is present in subclasses

Authorization sub-resource fields

If special discount or special product offering price or specific condition need an approval for ISP sale representative it is described here.

approver	A PartyRefOrPartyRoleRef. A reference to a Party or a PartyRole.
givenDate	A DateTime. Date when the authorization (approved or declined) was done.
name	A String. Name of the required authorization.
requestedDate	A DateTime. Date when the authorization is requested for.
signature	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
signatureRepresentation	A String. Indication that represents whether the signature is a physical paper signature or a digital signature.
state	A String. State of the authorization, such as: approved or declined.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BillingAccountRef sub-resource fields

BillingAccount reference. A BillingAccount is a detailed description of a bill structure.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
ratingType	A String. Indicates whether the account follows a specific payment option such as prepaid or postpaid.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'PortabilityOrder' resource object.


```
{
  "id": "999",
  "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/portabilityOrder/999",
  "@type": "PortabilityOrder",
  "@schemaLocation": "/schema-registry/schema/PortabilityOrder",
  "name": "Mobile Number Portability Order",
  "description": "Portability order for mobile number +61412345678",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "owner": "ExternalCRMSystem",
      "externalIdentifierType": "PortabilityReference",
      "id": "EXT-PORT-213321"
    }
  ],
  "category": "Mobile Number Portability",
  "state": "acknowledged",
  "creationDate": "2025-01-20T09:55:33.000Z",
  "requestedStartDate": "2025-01-22T08:00:00.000Z",
  "startDate": "2025-01-22T08:00:00.000Z",
```


© TM Forum 2026. All Rights Reserved.

```
  "requestedCompletionDate": "2025-01-23T17:00:00.000Z",
  "expectedCompletionDate": "2025-01-23T15:30:00.000Z",
  "completionDate": "2025-01-23T14:45:00.000Z",
  "note": [
    {
      "id": "1",
      "text": "Customer requested expedited processing for this portability order",
      "author": "John Doe",
      "date": "2025-01-20T10:15:22.000Z",
      "@type": "Note"
    }
  ],
  "groupRequestNumber": "GRP-41023",
  "portRequestNumber": "PRN-9988",
  "portRequestNumberVersion": "1.0",
  "dispute": false,
  "priority": "regular",
  "donor": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/partyRole/v5/partyRole/5646",
    "id": "5646",
    "name": "Notel Communications Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1234",
      "id": "1234",
      "name": "Notel Communications Inc",
      "@referredType": "Organization"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/partyRole/v5/partyRole/5647",
    "id": "5647",
    "name": "NewTel Services Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1235",
      "id": "1235",
      "name": "NewTel Services Pty Ltd",
      "@referredType": "Organization"
    }
  },
  "portabilitySubject": {
    "@type": "PortabilitySubject",
    "party": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56789",
      "id": "56789",
      "name": "Rahul K Sharma",
      "@referredType": "Individual"
    },
    "name": "Rahul K Sharma",
    "billingNamePrefix": "Mr.",
    "billingFirstName": "Rahul",
    "billingMiddleInitial": "K",
    "billingLastName": "Sharma",
    "billingNameSuffix": "",
```


© TM Forum 2026. All Rights Reserved.

```
    "authorization": {
      "@type": "Authorization",
      "name": "Mobile Number Portability Authorization",
      "givenDate": "2025-01-20T10:15:00.000Z",
      "requestedDate": "2025-01-20T10:15:00.000Z",
      "state": "approved",
      "signatureRepresentation": "digital",
      "approver": {
        "@type": "PartyRef",
        "@referredType": "Individual",
        "id": "56789",
        "name": "Rahul K Sharma"
      }
    },
    "address": {
      "@type": "GeographicAddress",
      "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/2121",
      "id": "2121",
      "externalIdentifier": [
        {
          "@type": "ExternalIdentifier",
          "owner": "ExternalCRMSystem",
          "externalIdentifierType": "CRMAddressID",
          "id": "2323"
        }
      ],
      "city": "Brighton",
      "country": "Australia",
      "locality": "Brighton",
      "postcode": "5048",
      "stateOrProvince": "SA",
      "streetName": "Strathmore",
      "streetNr": "225",
      "streetNrLast": "230",
      "streetNrLastSuffix": "B",
      "streetNrSuffix": "A",
      "streetSuffix": "Street",
      "streetType": "Street",
      "geographicLocation": {
        "@type": "GeographicLocationRef",
        "href": "https://mycsp.com:8080/tmf-
api/geographicLocation/v5/geographicLocation/67301845-ee43-4984-ba3b-b4fba4b98872",
        "id": "67301845-ee43-4984-ba3b-b4fba4b98872",
        "@referredType": "GeographicLocation"
      },
      "geographicSubAddress": [
        {
          "@type": "GeographicSubAddress",
          "id": "1",
          "buildingName": "Catalysts Building",
          "levelNumber": "3",
          "levelType": "floor",
          "name": "Suite 3A",
          "subUnit": [
            {
              "@type": "GeographicSubAddressUnit",
              "subUnitNumber": "7",
              "subUnitType": "unit"
            }
```


© TM Forum 2026. All Rights Reserved.

```
          ],
          "subAddressType": "commercial"
        }
      ]
    }
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56789",
        "id": "56789",
        "name": "Rahul K Sharma",
        "@referredType": "Individual"
      }
    }
  ],
  "orderRelationship": [
    {
      "@type": "OrderRelationship",
      "@schemaLocation": "/schema-registry/schema/OrderRelationship",
      "relationshipType": "dependency",
      "id": "123456",
      "href": "https://mycsp.com:8080/tmf-api/serviceOrdering/v5/serviceOrder/123456",
      "@referredType": "ServiceOrder"
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "state": "acknowledged",
      "@type": "PortabilityOrderItem",
      "action": "port",
      "portabilityActivationMode": "automatic",
      "portabilityEffectiveDate": "2025-01-22T08:00:00.000Z",
      "endUserName": "Rahul K Sharma",
      "resource": {
        "@type": "LogicalResource",
        "@baseType": "Resource",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventory/v5/resource/987654",
        "id": "987654",
        "name": "+61412345678",
        "description": "Mobile telephone number to be ported",
        "category": "TelephoneNumber",
        "startOperatingDate": "2025-01-22T08:00:00.000Z",
        "resourceCharacteristic": [
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "id": "1",
            "name": "NumberType",
            "value": "Mobile",
            "valueType": "string"
          },
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "id": "2",
```


© TM Forum 2026. All Rights Reserved.

```
            "name": "CountryCode",
            "value": "+61",
            "valueType": "string"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalog/v5/resourceSpecification/12345",
          "id": "12345",
          "name": "Mobile Number Specification",
          "@referredType": "ResourceSpecification",
          "version": "1.0"
        },
        "note": [
          {
            "@type": "Note",
            "id": "1",
            "author": "SystemAdmin",
            "date": "2025-01-20T09:55:33.000Z",
            "text": "Number currently active with donor provider"
          }
        ]
      },
      "note": [
        {
          "@type": "Note",
          "id": "1",
          "author": "OrderProcessor",
          "date": "2025-01-20T10:00:00.000Z",
          "text": "Order item validated and ready for processing"
        }
      ]
    }
  ]
}
```


CancelPortabilityOrder resource

Request for cancellation an existing portability order.

Resource model

© TM Forum 2026. All Rights Reserved.

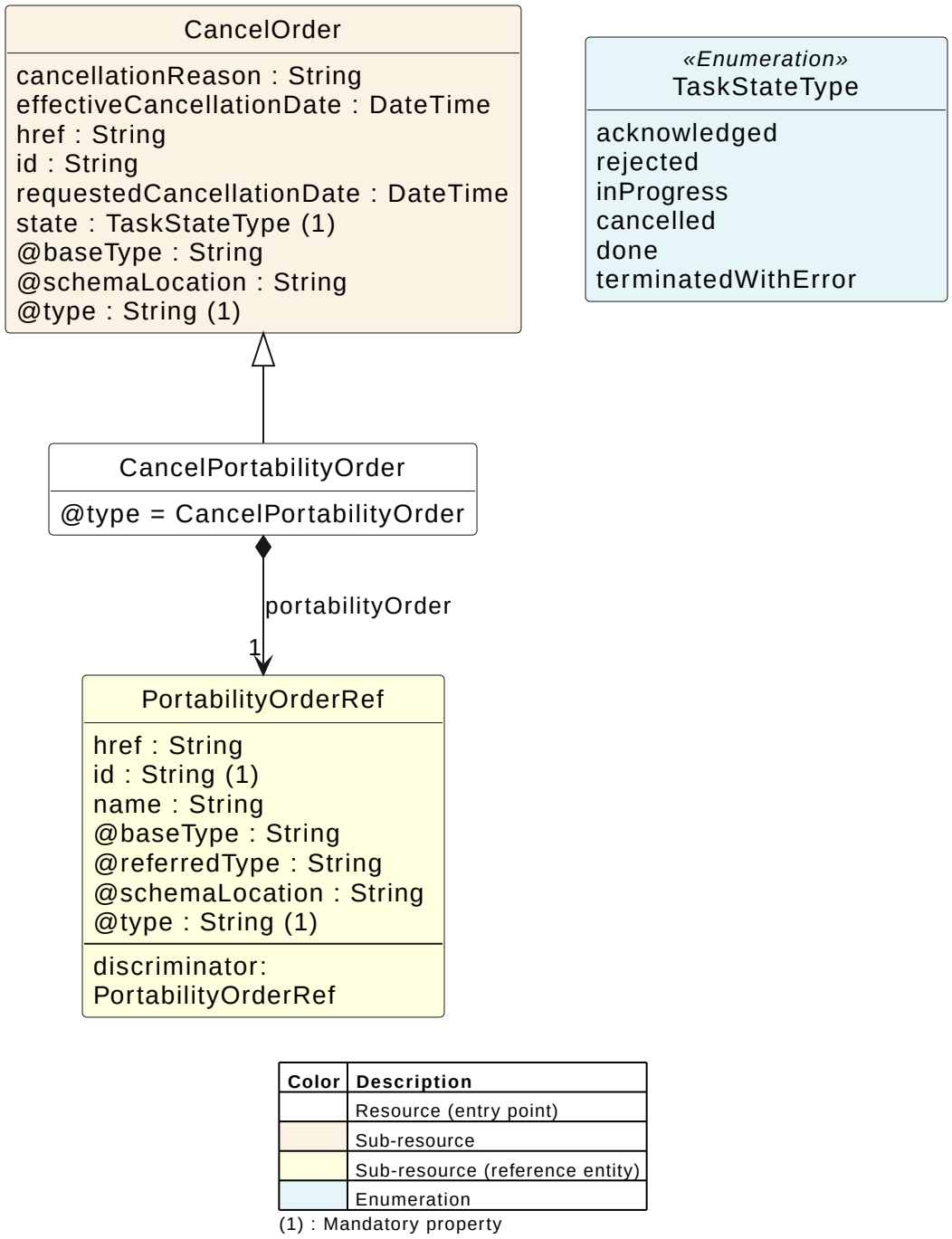


Figure 17 - CancelPortabilityOrder

Field descriptions

CancelPortabilityOrder fields

cancellationReason	A String. Reason why the order is cancelled.
effectiveCancellationDate	A DateTime. Date when the order is cancelled.

© TM Forum 2026. All Rights Reserved.

href	A String. Hyperlink reference.
id	A String. Unique identifier.
portabilityOrder	A PortabilityOrderRef [1]. The portability order which the reference is related with.
requestedCancellationDate	A DateTime. Date when the submitter wants the order to be cancelled.
state	A TaskStateType [1]. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PortabilityOrderRef sub-resource fields

The portability order which the reference is related with.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

© TM Forum 2026. All Rights Reserved.

We provide below a JSON representation as example of the 'CancelPortabilityOrder' resource object.


```
{
  "id": "CPO-001",
  "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CPO-001",
  "@type": "CancelPortabilityOrder",
  "requestedCancellationDate": "2025-01-21T14:30:00.000Z",
  "effectiveCancellationDate": "2025-01-21T15:00:00.000Z",
  "cancellationReason": "Customer request - service no longer required",
  "state": "done",
  "portabilityOrder": {
    "@type": "PortabilityOrderRef",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/999",
    "id": "999",
    "name": "Portability order for +61412345678",
    "@referredType": "PortabilityOrder"
  }
}
```


Notification Resource Models

7 notifications are defined for this API.

Notifications related to PortabilityOrder:

- Create Event
- Attribute Value Change Event
- Information Required Event
- State Change Event

Notifications related to CancelPortabilityOrder:

- Create Event
- Information Required Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name

© TM Forum 2026. All Rights Reserved.

as access field ("resourceName" placeholder).

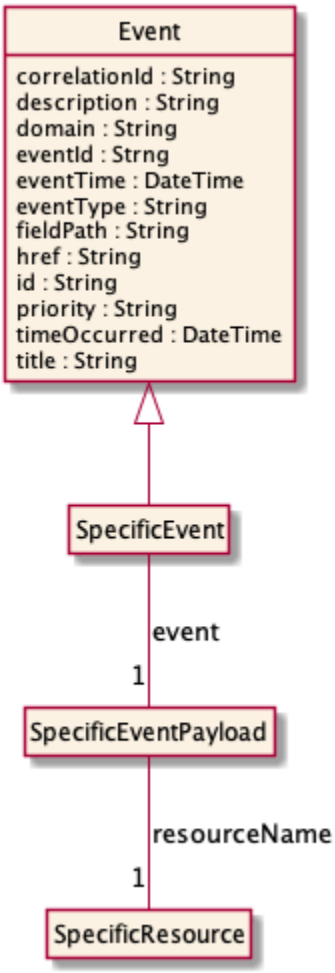


Figure 18 Notification Pattern

PortabilityOrder

Create Event

Notification sent when a new portability order is created

```
Content-Type: application/json
{
  "@type": "PortabilityOrderCreateEvent",
  "eventId": "CREATE-001",
  "eventTime": "2025-01-20T10:15:00.000Z",
  "eventType": "PortabilityOrderCreateEvent",
  "correlationId": "CORR-001",
  "domain": "resourcePortability",
  "title": "PortabilityOrder Create Event",
  "description": "A new portability order has been created",
  "priority": "regular",
  "event": {
```

© TM Forum 2026. All Rights Reserved.

```
    "portabilityOrder": {
      "id": "999",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/999",
      "state": "acknowledged",
      "creationDate": "2025-01-20T09:55:33.000Z",
      "description": "Portability order for mobile number +61412345678",
      "externalId": [
        {
          "id": "EXT-PORT-213321",
          "owner": "ExternalSystem",
          "externalIdentifierType": "PortabilityOrder",
          "@type": "ExternalIdentifier"
        }
      ],
      "requestedStartDate": "2025-01-22T08:00:00.000Z",
      "requestedCompletionDate": "2025-01-23T17:00:00.000Z",
      "expectedCompletionDate": "2025-01-23T15:30:00.000Z",
      "@type": "PortabilityOrder",
      "note": [
        {
          "id": "1",
          "text": "Customer requested expedited processing for this portability order",
          "author": "John Doe",
          "date": "2025-01-20T10:15:22.000Z",
          "@type": "Note"
        }
      ],
      "groupRequestNumber": "GRP-41023",
      "portRequestNumber": "PRN-9988",
      "portRequestNumberVersion": "1.0",
      "dispute": false,
      "priority": "expedited",
      "portabilitySubject": {
        "@type": "PortabilitySubject",
        "party": {
          "@type": "PartyRef",
          "id": "unknown",
          "name": "Rahul K Sharma",
          "@referredType": "Individual"
        },
        "name": "Rahul K Sharma",
        "billingNamePrefix": "Mr.",
        "billingFirstName": "Rahul",
        "billingMiddleInitial": "K",
        "billingLastName": "Sharma",
        "authorization": {
          "@type": "Authorization",
          "name": "Mobile Number Portability Authorization",
          "givenDate": "2025-01-20T10:15:00.000Z",
          "state": "approved",
          "signatureRepresentation": "digital",
          "approver": {
            "@type": "PartyRef",
            "@referredType": "Individual",
            "id": "56789",
            "name": "Rahul K Sharma"
          }
        },
        "address": {
```


© TM Forum 2026. All Rights Reserved.

```
          "@type": "GeographicAddress",
          "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/2121",
          "id": "2121",
          "city": "Brighton",
          "country": "Australia",
          "locality": "Brighton",
          "postcode": "5048",
          "stateOrProvince": "SA",
          "streetName": "Strathmore",
          "streetNr": "225",
          "streetType": "Street"
        }
      },
      "relatedParty": [
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "Customer",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/56789",
            "id": "56789",
            "name": "Rahul K Sharma",
            "@referredType": "Individual"
          }
        }
      ],
      "orderRelationship": [
        {
          "@type": "OrderRelationship",
          "relationshipType": "dependency",
          "id": "123456",
          "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/123456",
          "@referredType": "ServiceOrder"
        }
      ],
      "orderItem": [
        {
          "id": "1",
          "state": "acknowledged",
          "@type": "PortabilityOrderItem",
          "action": "port",
          "portabilityActivationMode": "automatic",
          "portabilityEffectiveDate": "2025-01-22T08:00:00.000Z",
          "endUserName": "Rahul K Sharma",
          "resource": {
            "@type": "LogicalResource",
            "@baseType": "Resource",
            "href": "https://mycsp.com:8080/tmf-
api/resourceInventory/v5/resource/987654",
            "id": "987654",
            "name": "+61412345678",
            "description": "Mobile telephone number to be ported",
            "category": "TelephoneNumber",
            "startOperatingDate": "2025-01-22T08:00:00.000Z",
            "resourceCharacteristic": [
              {
                "@type": "StringCharacteristic",
```


© TM Forum 2026. All Rights Reserved.

```
                "@baseType": "Characteristic",
                "id": "1",
                "name": "NumberType",
                "value": "Mobile",
                "valueType": "string"
              },
              {
                "@type": "StringCharacteristic",
                "@baseType": "Characteristic",
                "id": "2",
                "name": "CountryCode",
                "value": "+61",
                "valueType": "string"
              }
            ],
            "resourceSpecification": {
              "@type": "ResourceSpecificationRef",
              "href": "https://mycsp.com:8080/tmf-
api/resourceCatalog/v5/resourceSpecification/12345",
              "id": "12345",
              "name": "Mobile Number Specification",
              "@referredType": "ResourceSpecification",
              "version": "1.0"
            },
            "note": [
              {
                "@type": "Note",
                "id": "1",
                "author": "SystemAdmin",
                "date": "2025-01-20T09:55:33.000Z",
                "text": "Number currently active with donor provider"
              }
            ]
          }
        }
      ],
      "donor": {
        "@type": "ServiceProvider",
        "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5646",
        "id": "5646",
        "name": "Notel Communications Service Provider",
        "role": "ServiceProvider",
        "engagedParty": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/1234",
          "id": "1234",
          "name": "Notel Communications Inc",
          "@referredType": "Organization"
        }
      },
      "recipient": {
        "@type": "ServiceProvider",
        "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5647",
        "id": "5647",
        "name": "NewTel Services Service Provider",
        "role": "ServiceProvider",
        "engagedParty": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
```


© TM Forum 2026. All Rights Reserved.

```
api/partyManagement/v5/organization/1235",
          "id": "1235",
          "name": "NewTel Services Pty Ltd",
          "@referredType": "Organization"
        }
      }
    }
  }
}
```

Attribute Value Change Event

Notification sent when specific attribute values of a portability order change (excluding state changes)

```
Content-Type: application/json
{
  "@type": "PortabilityOrderAttributeValueChangeEvent",
  "eventId": "ATTR-CHANGE-001",
  "eventTime": "2025-01-22T08:30:00.000Z",
  "eventType": "PortabilityOrderAttributeValueChangeEvent",
  "correlationId": "CORR-002",
  "domain": "resourcePortability",
  "title": "PortabilityOrder Attribute Value Change Event",
  "description": "Portability order attributes have been updated",
  "priority": "regular",
  "event": {
    "portabilityOrder": {
      "@type": "PortabilityOrder",
      "state": "acknowledged",
      "creationDate": "2026-05-07T11:22:47Z",
      "priority": "regular",
      "donor": {
        "@type": "ServiceProvider",
        "name": "Vodafone UK",
        "engagedParty": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-
VODAFONE-UK-001",
          "id": "ORG-VODAFONE-UK-001",
          "name": "Vodafone UK",
          "@referredType": "Organization"
        }
      },
      "id": "PORT-2024-001",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/PORT-2024-001",
      "expectedCompletionDate": "2025-01-23T14:00:00.000Z",
      "note": [
        {
          "@type": "Note",
          "author": "System",
          "date": "2025-01-22T08:30:00.000Z",
          "text": "Port processing initiated with donor provider"
        }
      ]
```

© TM Forum 2026. All Rights Reserved.

```
    }
  }
}
```

Information Required Event

Notification sent when additional information is needed to process a portability order (e.g., missing authorization, incomplete customer details)

```
Content-Type: application/json
{
  "@type": "PortabilityOrderInformationRequiredEvent",
  "eventId": "INFO-REQ-PO-001",
  "eventTime": "2025-01-21T11:45:00.000Z",
  "eventType": "PortabilityOrderInformationRequiredEvent",
  "correlationId": "CORR-PO-002",
  "domain": "resourcePortability",
  "title": "PortabilityOrder Information Required Event",
  "description": "Additional information required to process portability order -
customer authorization signature missing",
  "priority": "expedited",
  "event": {
    "portabilityOrder": {
      "@type": "PortabilityOrderRef",
      "@referredType": "PortabilityOrder",
      "id": "PORT-2024-001",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/PORT-2024-001"
    },
    "informationRequired": {
      "id": "info-req-po-001",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/informationRequired/info-req-po-001",
      "@type": "InformationRequiredEvent",
      "informationRequiredPath": [
        {
          "op": "add",
          "path": "/portabilitySubject/party/authorization",
          "value": "Customer authorization signature is required. Please provide signed
authorization form."
        }
      ]
    }
  }
}
```

State Change Event

Notification sent when the state of a portability order changes (e.g., acknowledged to inProgress, inProgress to completed)

```
Content-Type: application/json
```

© TM Forum 2026. All Rights Reserved.

```
{
  "eventId": "STATE-CHANGE-PO-001",
  "@type": "PortabilityOrderStateChangeEvent",
  "eventTime": "2025-01-22T14:30:00.000Z",
  "eventType": "PortabilityOrderStateChangeEvent",
  "correlationId": "CORR-PO-001",
  "domain": "resourcePortability",
  "title": "PortabilityOrder State Change Event",
  "description": "Portability order state has changed from acknowledged to inProgress",
  "priority": "expedited",
  "event": {
    "portabilityOrder": {
      "@type": "PortabilityOrder",
      "id": "999",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/999",
      "state": "inProgress"
    }
  }
}
```


CancelPortabilityOrder

Create Event

Notification sent when a new cancel portability order request is created


```
Content-Type: application/json
{
  "@type": "CancelPortabilityOrderCreateEvent",
  "eventId": "CREATE-CPO-001",
  "eventTime": "2025-01-21T14:30:00.000Z",
  "eventType": "CancelPortabilityOrderCreateEvent",
  "correlationId": "CORR-007",
  "domain": "resourcePortability",
  "title": "CancelPortabilityOrder Create Event",
  "description": "A new cancel portability order has been created",
  "priority": "expedited",
  "event": {
    "cancelPortabilityOrder": {
      "id": "CPO-001",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CPO-001",
      "@type": "CancelPortabilityOrder",
      "requestedCancellationDate": "2025-01-21T14:30:00.000Z",
      "cancellationReason": "Customer request - service no longer required",
      "state": "acknowledged",
      "portabilityOrder": {
        "@type": "PortabilityOrderRef",
        "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/999",
        "id": "999",
        "name": "Portability order for +61412345678",
        "@referredType": "PortabilityOrder"
      }
    }
```


© TM Forum 2026. All Rights Reserved.

```
  }
}
```

Information Required Event

Notification sent when additional information is required for processing a cancel portability order request

```
Content-Type: application/json
{
  "@type": "CancelPortabilityOrderInformationRequiredEvent",
  "eventId": "INFO-REQ-001",
  "eventTime": "2025-01-21T14:45:00.000Z",
  "eventType": "CancelPortabilityOrderInformationRequiredEvent",
  "correlationId": "CORR-008",
  "domain": "resourcePortability",
  "title": "CancelPortabilityOrder Information Required Event",
  "description": "Additional information required for cancel portability order",
  "priority": "expedited",
  "event": {
    "cancelPortabilityOrder": {
      "@type": "CancelPortabilityOrderRef",
      "@referredType": "CancelPortabilityOrder",
      "id": "CPO-001",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CPO-001"
    },
    "informationRequired": {
      "id": "IR-2025-00123",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/informationRequired/IR-2025-00123",
      "@type": "InformationRequiredEvent",
      "informationRequiredPath": [
        {
          "op": "add",
          "path": "/cancellationReason",
          "value": "Cancellation reason is required"
        },
        {
          "op": "add",
          "path": "/supportingDocument",
          "value": "Supporting documentation is required"
        }
      ]
    }
  }
}
```

State Change Event

Notification sent when the state of a cancel portability order request changes

```
Content-Type: application/json
```

© TM Forum 2026. All Rights Reserved.

```
{
  "@type": "CancelPortabilityOrderStateChangeEvent",
  "eventId": "STATE-CHANGE-001",
  "eventTime": "2025-01-21T15:00:00.000Z",
  "eventType": "CancelPortabilityOrderStateChangeEvent",
  "correlationId": "CORR-003",
  "domain": "resourcePortability",
  "title": "CancelPortabilityOrder State Change Event",
  "description": "Cancel portability order state has changed",
  "priority": "regular",
  "event": {
    "cancelPortabilityOrder": {
      "@type": "CancelPortabilityOrder",
      "id": "CPO-001",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CPO-001",
      "state": "done"
    }
  }
}
```


© TM Forum 2026. All Rights Reserved.

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on PortabilityOrder

Retrieves a portabilityOrder by ID

`GET /portabilityOrder/{id}?fields=…`

Description

This operation retrieves a PortabilityOrder entity. Attribute selection is enabled for all first level attributes.

Usage samples

Retrieve a specific portability order using its unique identifier

Request

© TM Forum 2026. All Rights Reserved.

```
GET /portabilityOrder/PORT-2024-100
Content-Type: application/json
```

Response

```
200

{
  "id": "PORT-2024-100",
  "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/portabilityOrder/PORT-
2024-100",
  "state": "completed",
  "creationDate": "2025-01-18T11:20:00.000Z",
  "description": "Portability order for mobile number +61412345680",
  "externalId": [
    {
      "id": "EXT-PORT-213323",
      "owner": "ExternalSystem",
      "externalIdentifierType": "PortabilityOrder",
      "@type": "ExternalIdentifier"
    }
  ],
  "requestedStartDate": "2025-01-20T08:00:00.000Z",
  "requestedCompletionDate": "2025-01-21T17:00:00.000Z",
  "expectedCompletionDate": "2025-01-21T15:00:00.000Z",
  "completionDate": "2025-01-21T14:30:00.000Z",
  "@type": "PortabilityOrder",
  "note": [
    {
      "id": "1",
      "text": "Port completed ahead of schedule",
      "author": "System",
      "date": "2025-01-21T14:30:00.000Z",
      "@type": "Note"
    }
  ],
  "groupRequestNumber": "GRP-41025",
  "portRequestNumber": "PRN-9990",
  "portRequestNumberVersion": "1.0",
  "dispute": false,
  "priority": "regular",
  "portabilitySubject": {
    "@type": "PortabilitySubject",
    "party": {
      "@type": "PartyRef",
      "id": "unknown",
      "name": "Michael Johnson",
      "@referredType": "Individual"
    },
    "name": "Michael Johnson",
    "billingFirstName": "Michael",
    "billingLastName": "Johnson",
    "authorization": {
      "@type": "Authorization",
      "name": "Mobile Number Portability Authorization",
      "givenDate": "2025-01-18T11:45:00.000Z",
```

© TM Forum 2026. All Rights Reserved.

```
      "state": "approved",
      "signatureRepresentation": "digital"
    },
    "address": {
      "@type": "GeographicAddress",
      "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/2122",
      "id": "2122",
      "city": "Adelaide",
      "country": "Australia",
      "postcode": "5000",
      "stateOrProvince": "SA",
      "streetName": "King William",
      "streetNr": "100",
      "streetType": "Street"
    }
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56790",
        "id": "56790",
        "name": "Michael Johnson",
        "@referredType": "Individual"
      }
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "state": "confirmed",
      "@type": "PortabilityOrderItem",
      "action": "port",
      "portabilityActivationMode": "automatic",
      "portabilityEffectiveDate": "2025-01-21T14:30:00.000Z",
      "endUserName": "Michael Johnson",
      "resource": {
        "@type": "LogicalResource",
        "@baseType": "Resource",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventory/v5/resource/987655",
        "id": "987655",
        "name": "+61412345680",
        "description": "Mobile telephone number successfully ported",
        "category": "TelephoneNumber",
        "startOperatingDate": "2025-01-21T14:30:00.000Z",
        "resourceCharacteristic": [
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "id": "1",
            "name": "NumberType",
            "value": "Mobile",
            "valueType": "string"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
```


© TM Forum 2026. All Rights Reserved.

```
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalog/v5/resourceSpecification/12345",
          "id": "12345",
          "name": "Mobile Number Specification",
          "@referredType": "ResourceSpecification",
          "version": "1.0"
        }
      }
    }
  ],
  "donor": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5646",
    "id": "5646",
    "name": "Notel Communications Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1234",
      "id": "1234",
      "name": "Notel Communications Inc",
      "@referredType": "Organization"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5647",
    "id": "5647",
    "name": "NewTel Services Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1235",
      "id": "1235",
      "name": "NewTel Services Pty Ltd",
      "@referredType": "Organization"
    }
  }
}
```


List or find portabilityOrder objects

`GET /portabilityOrder?fields=…&{filtering}`

Description

This operation list PortabilityOrder entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of portability orders with date range filtering, sorting, and pagination

© TM Forum 2026. All Rights Reserved.

Request

```
GET /portabilityOrder?offset=0&limit=50&before=2025-01-01T00:00:00Z&after=2024-12-
01T00:00:00Z&sort=creationDate:desc&filter=state=inProgress,priority=expedited
Content-Type: application/json
```

Response

```
200

[
  {
    "id": "999",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/999",
    "state": "acknowledged",
    "creationDate": "2025-01-20T09:55:33.000Z",
    "description": "Portability order for mobile number +61412345678",
    "externalId": [
      {
        "id": "EXT-PORT-213321",
        "owner": "ExternalSystem",
        "externalIdentifierType": "PortabilityOrder",
        "@type": "ExternalIdentifier"
      }
    ],
    "requestedStartDate": "2025-01-22T08:00:00.000Z",
    "requestedCompletionDate": "2025-01-23T17:00:00.000Z",
    "@type": "PortabilityOrder",
    "priority": "expedited",
    "orderItem": [
      {
        "@type": "PortabilityOrderItem",
        "id": "POI-2026-00101"
      }
    ],
    "donor": {
      "@type": "ServiceProvider",
      "id": "SP-DONOR-001",
      "name": "TelecomOne",
      "engagedParty": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-
001",
        "id": "ORG-001",
        "name": "TelecomOne Ltd",
        "@referredType": "Organization"
      }
    },
    "recipient": {
      "@type": "ServiceProvider",
      "id": "SP-RECIPIENT-001",
      "name": "UltraTel",
      "engagedParty": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-
```

© TM Forum 2026. All Rights Reserved.

```
002",
        "id": "ORG-002",
        "name": "UltraTel Corp",
        "@referredType": "Organization"
      }
    },
    "portabilitySubject": {
      "@type": "PortabilitySubject",
      "party": {
        "@type": "PartyRef",
        "id": "CUST-001",
        "name": "Rahul K Sharma",
        "@referredType": "Individual"
      },
      "name": "Rahul K Sharma"
    }
  },
  {
    "id": "998",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/998",
    "state": "inProgress",
    "creationDate": "2025-01-19T14:30:00.000Z",
    "description": "Portability order for mobile number +61412345677",
    "externalId": [
      {
        "id": "EXT-PORT-213320",
        "owner": "ExternalSystem",
        "externalIdentifierType": "PortabilityOrder",
        "@type": "ExternalIdentifier"
      }
    ],
    "requestedStartDate": "2025-01-21T08:00:00.000Z",
    "requestedCompletionDate": "2025-01-22T17:00:00.000Z",
    "@type": "PortabilityOrder",
    "priority": "regular",
    "orderItem": [
      {
        "@type": "PortabilityOrderItem",
        "id": "POI-2026-00102"
      }
    ],
    "donor": {
      "@type": "ServiceProvider",
      "id": "SP-DONOR-003",
      "name": "TelecomOne",
      "engagedParty": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-
003",
        "id": "ORG-003",
        "name": "TelecomTwo Ltd",
        "@referredType": "Organization"
      }
    },
    "recipient": {
      "@type": "ServiceProvider",
      "id": "SP-RECIPIENT-002",
      "name": "UltraTel",
      "engagedParty": {
```


© TM Forum 2026. All Rights Reserved.

```
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-
004",
        "id": "ORG-004",
        "name": "UltraTelTwo Corp",
        "@referredType": "Organization"
      }
    },
    "portabilitySubject": {
      "@type": "PortabilitySubject",
      "party": {
        "@type": "PartyRef",
        "id": "CUST-002",
        "name": "Jonathan Sharma",
        "@referredType": "Individual"
      },
      "name": "Jonathan Sharma"
    }
  }
]
```

Creates a portabilityOrder

`POST /portabilityOrder?fields=…&{filtering}`

Description

This operation creates a PortabilityOrder entity.

Mandatory Attributes

Mandatory Attributes	Rule
donor	M
orderItem	M
portabilitySubject	M
priority	M
recipient	M
@type	

Usage samples

Create a new portability order for resource portability with order items and related parties

Request

```
POST /portabilityOrder
Content-Type: application/json
```

© TM Forum 2026. All Rights Reserved.

```
{
  "description": "Portability order for mobile number +61412345678",
  "externalId": [
    {
      "id": "EXT-PORT-213321",
      "owner": "ExternalSystem",
      "externalIdentifierType": "PortabilityOrder",
      "@type": "ExternalIdentifier"
    }
  ],
  "requestedStartDate": "2025-01-22T08:00:00.000Z",
  "requestedCompletionDate": "2025-01-23T17:00:00.000Z",
  "@type": "PortabilityOrder",
  "note": [
    {
      "text": "Customer requested expedited processing for this portability order",
      "author": "John Doe",
      "date": "2025-01-20T10:15:22.000Z",
      "@type": "Note"
    }
  ],
  "groupRequestNumber": "GRP-41023",
  "portRequestNumber": "PRN-9988",
  "portRequestNumberVersion": "1.0",
  "dispute": false,
  "priority": "expedited",
  "donor": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/partyRole/v5/partyRole/5646",
    "id": "5646",
    "name": "Notel Communications Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1234",
      "id": "1234",
      "name": "Notel Communications Inc",
      "@referredType": "Organization"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/partyRole/v5/partyRole/5647",
    "id": "5647",
    "name": "NewTel Services Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1235",
      "id": "1235",
      "name": "NewTel Services Pty Ltd",
      "@referredType": "Organization"
    }
  },
  "portabilitySubject": {
    "@type": "PortabilitySubject",
    "party": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56789",
```


© TM Forum 2026. All Rights Reserved.

```
      "id": "56789",
      "name": "Rahul K Sharma",
      "@referredType": "Individual"
    },
    "name": "Rahul K Sharma",
    "billingNamePrefix": "Mr.",
    "billingFirstName": "Rahul",
    "billingMiddleInitial": "K",
    "billingLastName": "Sharma",
    "authorization": {
      "@type": "Authorization",
      "name": "Mobile Number Portability Authorization",
      "givenDate": "2025-01-20T09:30:00.000Z",
      "requestedDate": "2025-01-20T09:30:00.000Z",
      "signatureRepresentation": "digital",
      "approver": {
        "@type": "PartyRef",
        "@referredType": "Individual",
        "id": "56789",
        "name": "Rahul K Sharma"
      }
    },
    "address": {
      "@type": "GeographicAddress",
      "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/2121",
      "id": "2121",
      "city": "Brighton",
      "country": "Australia",
      "locality": "Brighton",
      "postcode": "5048",
      "stateOrProvince": "SA",
      "streetName": "Strathmore",
      "streetNr": "225",
      "streetType": "Street"
    }
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56789",
        "id": "56789",
        "name": "Rahul K Sharma",
        "@referredType": "Individual"
      }
    }
  ],
  "orderItem": [
    {
      "@type": "PortabilityOrderItem",
      "action": "port",
      "portabilityActivationMode": "automatic",
      "portabilityEffectiveDate": "2025-01-22T08:00:00.000Z",
      "endUserName": "Rahul K Sharma",
      "resource": {
        "@type": "LogicalResource",
        "@baseType": "Resource",
```


© TM Forum 2026. All Rights Reserved.

```
        "name": "+61412345678",
        "description": "Mobile telephone number to be ported",
        "category": "TelephoneNumber",
        "resourceCharacteristic": [
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "NumberType",
            "value": "Mobile",
            "valueType": "string"
          },
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "name": "CountryCode",
            "value": "+61",
            "valueType": "string"
          }
        ]
      },
      "id": "1"
    }
  ]
}
```

Response

```
201

{
  "id": "999",
  "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/portabilityOrder/999",
  "state": "acknowledged",
  "creationDate": "2025-01-20T09:55:33.000Z",
  "description": "Portability order for mobile number +61412345678",
  "externalId": [
    {
      "id": "EXT-PORT-213321",
      "owner": "ExternalSystem",
      "externalIdentifierType": "PortabilityOrder",
      "@type": "ExternalIdentifier"
    }
  ],
  "requestedStartDate": "2025-01-22T08:00:00.000Z",
  "requestedCompletionDate": "2025-01-23T17:00:00.000Z",
  "expectedCompletionDate": "2025-01-23T15:30:00.000Z",
  "@type": "PortabilityOrder",
  "note": [
    {
      "id": "1",
      "text": "Customer requested expedited processing for this portability order",
      "author": "John Doe",
      "date": "2025-01-20T10:15:22.000Z",
      "@type": "Note"
    }
  ],
  "groupRequestNumber": "GRP-41023",
```

© TM Forum 2026. All Rights Reserved.

```
  "portRequestNumber": "PRN-9988",
  "portRequestNumberVersion": "1.0",
  "dispute": false,
  "priority": "expedited",
  "donor": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/partyRole/v5/partyRole/5646",
    "id": "5646",
    "name": "Notel Communications Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1234",
      "id": "1234",
      "name": "Notel Communications Inc",
      "@referredType": "Organization"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/partyRole/v5/partyRole/5647",
    "id": "5647",
    "name": "NewTel Services Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1235",
      "id": "1235",
      "name": "NewTel Services Pty Ltd",
      "@referredType": "Organization"
    }
  },
  "portabilitySubject": {
    "@type": "PortabilitySubject",
    "party": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56789",
      "id": "56789",
      "name": "Rahul K Sharma",
      "@referredType": "Individual"
    },
    "name": "Rahul K Sharma",
    "billingNamePrefix": "Mr.",
    "billingFirstName": "Rahul",
    "billingMiddleInitial": "K",
    "billingLastName": "Sharma",
    "authorization": {
      "@type": "Authorization",
      "name": "Mobile Number Portability Authorization",
      "givenDate": "2025-01-20T09:30:00.000Z",
      "requestedDate": "2025-01-20T09:30:00.000Z",
      "state": "approved",
      "signatureRepresentation": "digital",
      "approver": {
        "@type": "PartyRef",
        "@referredType": "Individual",
        "id": "56789",
        "name": "Rahul K Sharma"
      }
    },
```


© TM Forum 2026. All Rights Reserved.

```
    "address": {
      "@type": "GeographicAddress",
      "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/2121",
      "id": "2121",
      "city": "Brighton",
      "country": "Australia",
      "locality": "Brighton",
      "postcode": "5048",
      "stateOrProvince": "SA",
      "streetName": "Strathmore",
      "streetNr": "225",
      "streetType": "Street"
    }
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56789",
        "id": "56789",
        "name": "Rahul K Sharma",
        "@referredType": "Individual"
      }
    }
  ],
  "orderRelationship": [
    {
      "@type": "OrderRelationship",
      "relationshipType": "dependency",
      "id": "123456",
      "href": "https://mycsp.com:8080/tmf-api/serviceOrdering/v5/serviceOrder/123456",
      "@referredType": "ServiceOrder"
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "state": "acknowledged",
      "@type": "PortabilityOrderItem",
      "action": "port",
      "portabilityActivationMode": "automatic",
      "portabilityEffectiveDate": "2025-01-22T08:00:00.000Z",
      "endUserName": "Rahul K Sharma",
      "resource": {
        "@type": "LogicalResource",
        "@baseType": "Resource",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventory/v5/resource/987654",
        "id": "987654",
        "name": "+61412345678",
        "description": "Mobile telephone number to be ported",
        "category": "TelephoneNumber",
        "startOperatingDate": "2025-01-22T08:00:00.000Z",
        "resourceCharacteristic": [
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "id": "1",
```


© TM Forum 2026. All Rights Reserved.

```
            "name": "NumberType",
            "value": "Mobile",
            "valueType": "string"
          },
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "id": "2",
            "name": "CountryCode",
            "value": "+61",
            "valueType": "string"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalog/v5/resourceSpecification/12345",
          "id": "12345",
          "name": "Mobile Number Specification",
          "@referredType": "ResourceSpecification",
          "version": "1.0"
        },
        "note": [
          {
            "@type": "Note",
            "id": "1",
            "author": "SystemAdmin",
            "date": "2025-01-20T09:55:33.000Z",
            "text": "Number currently active with donor provider"
          }
        ]
      }
    }
  ]
}
```


Updates partially a portabilityOrder

`PATCH /portabilityOrder/{id}?fields=…`

Description

This operation allows partial updates of a PortabilityOrder entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

© TM Forum 2026. All Rights Reserved.

Non Patchable Attributes	Rule
creationDate	
href	
id	
state	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
cancellationDate	
cancellationReason	
category	
completionDate	
description	
dispute	
donor	
expectedCompletionDate	
externalId	
groupRequestNumber	
lastUpdate	
name	
note	
orderItem	
orderRelationship	
portabilitySubject	
portRequestNumber	
portRequestNumberVersion	
priority	
recipient	

© TM Forum 2026. All Rights Reserved.

Patchable Attributes	Rule
relatedParty	
requestedCompletionDate	
requestedStartDate	
startDate	

Usage samples

Update a portability order using RFC 7396 JSON Merge Patch for simple field updates

Request

```
PATCH /portabilityOrder/PORT-2024-001
Content-Type: application/merge-patch+json

{
  "@type": "PortabilityOrder",
  "expectedCompletionDate": "2025-01-23T14:00:00.000Z",
  "priority": "expedited",
  "orderItem": [
    {
      "@type": "PortabilityOrderItem",
      "id": "POI-2026-00101"
    }
  ],
  "donor": {
    "@type": "ServiceProvider",
    "id": "SP-DONOR-001",
    "name": "TelecomOne",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-001",
      "id": "ORG-001",
      "name": "TelecomOne Ltd",
      "@referredType": "Organization"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "id": "SP-RECIPIENT-001",
    "name": "UltraTel",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-002",
      "id": "ORG-002",
      "@referredType": "Organization"
    }
  },
  "note": [
    {
      "@type": "Note",
      "author": "System",
```

© TM Forum 2026. All Rights Reserved.

```
      "date": "2025-01-22T08:30:00.000Z",
      "text": "Port processing initiated with donor provider"
    }
  ]
}
```

Response

```
200

{
  "id": "PORT-2024-001",
  "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/portabilityOrder/PORT-
2024-001",
  "state": "inProgress",
  "creationDate": "2025-01-20T11:00:00.000Z",
  "description": "Portability order for mobile number +61412345681",
  "externalId": [
    {
      "id": "EXT-PORT-213324",
      "owner": "ExternalSystem",
      "externalIdentifierType": "PortabilityOrder",
      "@type": "ExternalIdentifier"
    }
  ],
  "requestedStartDate": "2025-01-22T08:00:00.000Z",
  "requestedCompletionDate": "2025-01-23T17:00:00.000Z",
  "expectedCompletionDate": "2025-01-23T14:00:00.000Z",
  "@type": "PortabilityOrder",
  "note": [
    {
      "id": "1",
      "text": "Customer requested expedited processing",
      "author": "John Doe",
      "date": "2025-01-20T11:15:00.000Z",
      "@type": "Note"
    },
    {
      "id": "2",
      "text": "Port processing initiated with donor provider",
      "author": "System",
      "date": "2025-01-22T08:30:00.000Z",
      "@type": "Note"
    }
  ],
  "groupRequestNumber": "GRP-41026",
  "portRequestNumber": "PRN-9991",
  "portRequestNumberVersion": "1.0",
  "dispute": false,
  "priority": "expedited",
  "portabilitySubject": {
    "@type": "PortabilitySubject",
    "party": {
      "@type": "PartyRef",
      "id": "unknown",
      "name": "Emily Brown",
      "@referredType": "Individual"
```

© TM Forum 2026. All Rights Reserved.

```
    },
    "name": "Emily Brown",
    "billingFirstName": "Emily",
    "billingLastName": "Brown",
    "authorization": {
      "@type": "Authorization",
      "name": "Mobile Number Portability Authorization",
      "givenDate": "2025-01-20T11:30:00.000Z",
      "state": "approved",
      "signatureRepresentation": "digital"
    },
    "address": {
      "@type": "GeographicAddress",
      "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/2123",
      "id": "2123",
      "city": "Melbourne",
      "country": "Australia",
      "postcode": "3000",
      "stateOrProvince": "VIC",
      "streetName": "Collins",
      "streetNr": "500",
      "streetType": "Street"
    }
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56792",
        "id": "56792",
        "name": "Emily Brown",
        "@referredType": "Individual"
      }
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "state": "confirmed",
      "@type": "PortabilityOrderItem",
      "action": "port",
      "portabilityActivationMode": "automatic",
      "portabilityEffectiveDate": "2025-01-22T08:00:00.000Z",
      "endUserName": "Emily Brown",
      "resource": {
        "@type": "LogicalResource",
        "@baseType": "Resource",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventory/v5/resource/987656",
        "id": "987656",
        "name": "+61412345681",
        "description": "Mobile telephone number being ported",
        "category": "TelephoneNumber",
        "resourceCharacteristic": [
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "id": "1",
```


© TM Forum 2026. All Rights Reserved.

```
            "name": "NumberType",
            "value": "Mobile",
            "valueType": "string"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalog/v5/resourceSpecification/12345",
          "id": "12345",
          "name": "Mobile Number Specification",
          "@referredType": "ResourceSpecification",
          "version": "1.0"
        }
      }
    }
  ],
  "donor": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5646",
    "id": "5646",
    "name": "Notel Communications Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1234",
      "id": "1234",
      "name": "Notel Communications Inc",
      "@referredType": "Organization"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5647",
    "id": "5647",
    "name": "NewTel Services Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1235",
      "id": "1235",
      "name": "NewTel Services Pty Ltd",
      "@referredType": "Organization"
    }
  }
}
```

Update a portability order using JSON Patch with query capabilities for complex updates

Request

```
PATCH /portabilityOrder/PORT-2024-002
Content-Type: application/json-patch-query+json

[
  {
    "op": "add",
```

© TM Forum 2026. All Rights Reserved.

```
    "path": "/orderItem[?(@.id=='1')]/note/-",
    "value": {
      "@type": "Note",
      "author": "System",
      "date": "2025-01-22T16:30:00.000Z",
      "text": "Port failed - donor provider rejected the request"
    }
  },
  {
    "op": "replace",
    "path": "/stateReason",
    "value": "Donor Rejection"
  }
]
```

Response

```
200

{
  "id": "PORT-2024-002",
  "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/portabilityOrder/PORT-
2024-002",
  "state": "failed",
  "creationDate": "2025-01-20T13:00:00.000Z",
  "description": "Portability order for mobile number +61412345683",
  "externalId": [
    {
      "id": "EXT-PORT-213326",
      "owner": "ExternalSystem",
      "externalIdentifierType": "PortabilityOrder",
      "@type": "ExternalIdentifier"
    }
  ],
  "requestedStartDate": "2025-01-22T08:00:00.000Z",
  "requestedCompletionDate": "2025-01-23T17:00:00.000Z",
  "@type": "PortabilityOrder",
  "note": [
    {
      "id": "1",
      "text": "Order created for customer",
      "author": "Jane Smith",
      "date": "2025-01-20T13:15:00.000Z",
      "@type": "Note"
    }
  ],
  "groupRequestNumber": "GRP-41028",
  "portRequestNumber": "PRN-9993",
  "portRequestNumberVersion": "1.0",
  "dispute": false,
  "priority": "regular",
  "portabilitySubject": {
    "@type": "PortabilitySubject",
    "party": {
      "@type": "PartyRef",
      "id": "unknown",
      "name": "Robert Taylor",
```

© TM Forum 2026. All Rights Reserved.

```
      "@referredType": "Individual"
    },
    "name": "Robert Taylor",
    "billingFirstName": "Robert",
    "billingLastName": "Taylor",
    "authorization": {
      "@type": "Authorization",
      "name": "Mobile Number Portability Authorization",
      "givenDate": "2025-01-20T13:30:00.000Z",
      "state": "approved",
      "signatureRepresentation": "digital"
    },
    "address": {
      "@type": "GeographicAddress",
      "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/2125",
      "id": "2125",
      "city": "Brisbane",
      "country": "Australia",
      "postcode": "4000",
      "stateOrProvince": "QLD",
      "streetName": "Queen",
      "streetNr": "300",
      "streetType": "Street"
    }
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56794",
        "id": "56794",
        "name": "Robert Taylor",
        "@referredType": "Individual"
      }
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "state": "confirmed",
      "@type": "PortabilityOrderItem",
      "action": "port",
      "portabilityActivationMode": "automatic",
      "portabilityEffectiveDate": "2025-01-22T08:00:00.000Z",
      "endUserName": "Robert Taylor",
      "note": [
        {
          "id": "1",
          "@type": "Note",
          "author": "System",
          "date": "2025-01-22T16:30:00.000Z",
          "text": "Port failed - donor provider rejected the request"
        }
      ],
      "resource": {
        "@type": "LogicalResource",
        "@baseType": "Resource",
```


© TM Forum 2026. All Rights Reserved.

```
        "href": "https://mycsp.com:8080/tmf-api/resourceInventory/v5/resource/987658",
        "id": "987658",
        "name": "+61412345683",
        "description": "Mobile telephone number - port failed",
        "category": "TelephoneNumber",
        "resourceCharacteristic": [
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "id": "1",
            "name": "NumberType",
            "value": "Mobile",
            "valueType": "string"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalog/v5/resourceSpecification/12345",
          "id": "12345",
          "name": "Mobile Number Specification",
          "@referredType": "ResourceSpecification",
          "version": "1.0"
        }
      }
    }
  ],
  "donor": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5646",
    "id": "5646",
    "name": "Notel Communications Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1234",
      "id": "1234",
      "name": "Notel Communications Inc",
      "@referredType": "Organization"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5647",
    "id": "5647",
    "name": "NewTel Services Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1235",
      "id": "1235",
      "name": "NewTel Services Pty Ltd",
      "@referredType": "Organization"
    }
  }
}
```


Update a portability order using RFC 6902 JSON Patch operations (add, remove, replace)

© TM Forum 2026. All Rights Reserved.

Request

```
PATCH /portabilityOrder/PORT-2024-003
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/stateReason",
    "value": "Port Completed"
  },
  {
    "op": "replace",
    "path": "/completionDate",
    "value": "2025-01-23T14:45:00.000Z"
  },
  {
    "op": "add",
    "path": "/note/-",
    "value": {
      "@type": "Note",
      "author": "System",
      "date": "2025-01-23T14:45:00.000Z",
      "text": "Number portability completed successfully"
    }
  }
]
```

Response

```
200

{
  "id": "PORT-2024-003",
  "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/portabilityOrder/PORT-
2024-003",
  "state": "completed",
  "creationDate": "2025-01-20T12:00:00.000Z",
  "description": "Portability order for mobile number +61412345682",
  "externalId": [
    {
      "id": "EXT-PORT-213325",
      "owner": "ExternalSystem",
      "externalIdentifierType": "PortabilityOrder",
      "@type": "ExternalIdentifier"
    }
  ],
  "requestedStartDate": "2025-01-22T08:00:00.000Z",
  "requestedCompletionDate": "2025-01-23T17:00:00.000Z",
  "expectedCompletionDate": "2025-01-23T15:00:00.000Z",
  "completionDate": "2025-01-23T14:45:00.000Z",
  "@type": "PortabilityOrder",
  "note": [
    {
      "id": "1",
      "text": "Initial order created",
```

© TM Forum 2026. All Rights Reserved.

```
      "author": "John Doe",
      "date": "2025-01-20T12:15:00.000Z",
      "@type": "Note"
    },
    {
      "id": "2",
      "text": "Number portability completed successfully",
      "author": "System",
      "date": "2025-01-23T14:45:00.000Z",
      "@type": "Note"
    }
  ],
  "groupRequestNumber": "GRP-41027",
  "portRequestNumber": "PRN-9992",
  "portRequestNumberVersion": "1.0",
  "dispute": false,
  "priority": "regular",
  "portabilitySubject": {
    "@type": "PortabilitySubject",
    "party": {
      "@type": "PartyRef",
      "id": "unknown",
      "name": "David Lee",
      "@referredType": "Individual"
    },
    "name": "David Lee",
    "billingFirstName": "David",
    "billingLastName": "Lee",
    "authorization": {
      "@type": "Authorization",
      "name": "Mobile Number Portability Authorization",
      "givenDate": "2025-01-20T12:30:00.000Z",
      "state": "approved",
      "signatureRepresentation": "digital"
    },
    "address": {
      "@type": "GeographicAddress",
      "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/2124",
      "id": "2124",
      "city": "Sydney",
      "country": "Australia",
      "postcode": "2000",
      "stateOrProvince": "NSW",
      "streetName": "George",
      "streetNr": "200",
      "streetType": "Street"
    }
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56793",
        "id": "56793",
        "name": "David Lee",
        "@referredType": "Individual"
      }
```


© TM Forum 2026. All Rights Reserved.

```
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "state": "confirmed",
      "@type": "PortabilityOrderItem",
      "action": "port",
      "portabilityActivationMode": "automatic",
      "portabilityEffectiveDate": "2025-01-23T14:45:00.000Z",
      "endUserName": "David Lee",
      "resource": {
        "@type": "LogicalResource",
        "@baseType": "Resource",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventory/v5/resource/987657",
        "id": "987657",
        "name": "+61412345682",
        "description": "Mobile telephone number successfully ported",
        "category": "TelephoneNumber",
        "startOperatingDate": "2025-01-23T14:45:00.000Z",
        "resourceCharacteristic": [
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "id": "1",
            "name": "NumberType",
            "value": "Mobile",
            "valueType": "string"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalog/v5/resourceSpecification/12345",
          "id": "12345",
          "name": "Mobile Number Specification",
          "@referredType": "ResourceSpecification",
          "version": "1.0"
        }
      }
    }
  ],
  "donor": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5646",
    "id": "5646",
    "name": "Notel Communications Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1234",
      "id": "1234",
      "name": "Notel Communications Inc",
      "@referredType": "Organization"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5647",
    "id": "5647",
```


© TM Forum 2026. All Rights Reserved.

```
    "name": "NewTel Services Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1235",
      "id": "1235",
      "name": "NewTel Services Pty Ltd",
      "@referredType": "Organization"
    }
  }
}
```

Partially update an existing portability order using standard JSON merge patch

Request

```
PATCH /portabilityOrder/PORT-2024-004
Content-Type: application/json

{
  "@type": "PortabilityOrder",
  "priority": "expedited",
  "orderItem": [
    {
      "@type": "PortabilityOrderItem",
      "id": "POI-2026-00101"
    }
  ],
  "donor": {
    "@type": "ServiceProvider",
    "id": "SP-DONOR-001",
    "name": "TelecomOne",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-001",
      "id": "ORG-001",
      "name": "TelecomOne Ltd",
      "@referredType": "Organization"
    }
  },
  "recipient": {
    "@type": "ServiceProvider",
    "id": "SP-RECIPIENT-001",
    "name": "UltraTel",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-002",
      "id": "ORG-002",
      "@referredType": "Organization"
    }
  },
  "note": [
    {
      "@type": "Note",
      "author": "Customer Service",
      "date": "2025-01-21T11:00:00.000Z",
      "text": "Customer requested cancellation"
```

© TM Forum 2026. All Rights Reserved.

```
    }
  ]
}
```

Response

```
200

{
  "id": "PORT-2024-004",
  "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/portabilityOrder/PORT-
2024-004",
  "state": "cancelled",
  "creationDate": "2025-01-20T14:00:00.000Z",
  "recipient": {
    "@type": "ServiceProvider",
    "id": "SP-RECIPIENT-001",
    "name": "UltraTel",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/ORG-002",
      "id": "ORG-002",
      "name": "UltraTel Corp",
      "@referredType": "Organization"
    }
  },
  "description": "Portability order for mobile number +61412345684",
  "externalId": [
    {
      "id": "EXT-PORT-213327",
      "owner": "ExternalSystem",
      "externalIdentifierType": "PortabilityOrder",
      "@type": "ExternalIdentifier"
    }
  ],
  "requestedStartDate": "2025-01-22T08:00:00.000Z",
  "requestedCompletionDate": "2025-01-23T17:00:00.000Z",
  "@type": "PortabilityOrder",
  "note": [
    {
      "id": "1",
      "text": "Initial order created",
      "author": "System",
      "date": "2025-01-20T14:15:00.000Z",
      "@type": "Note"
    },
    {
      "id": "2",
      "text": "Customer requested cancellation",
      "author": "Customer Service",
      "date": "2025-01-21T11:00:00.000Z",
      "@type": "Note"
    }
  ],
  "groupRequestNumber": "GRP-41029",
  "portRequestNumber": "PRN-9994",
  "portRequestNumberVersion": "1.0",
```

© TM Forum 2026. All Rights Reserved.

```
  "dispute": false,
  "priority": "regular",
  "portabilitySubject": {
    "@type": "PortabilitySubject",
    "party": {
      "@type": "PartyRef",
      "id": "unknown",
      "name": "Lisa Anderson",
      "@referredType": "Individual"
    },
    "name": "Lisa Anderson",
    "billingFirstName": "Lisa",
    "billingLastName": "Anderson",
    "authorization": {
      "@type": "Authorization",
      "name": "Mobile Number Portability Authorization",
      "givenDate": "2025-01-20T14:30:00.000Z",
      "state": "approved",
      "signatureRepresentation": "digital"
    },
    "address": {
      "@type": "GeographicAddress",
      "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/2126",
      "id": "2126",
      "city": "Perth",
      "country": "Australia",
      "postcode": "6000",
      "stateOrProvince": "WA",
      "streetName": "Murray",
      "streetNr": "400",
      "streetType": "Street"
    }
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/56795",
        "id": "56795",
        "name": "Lisa Anderson",
        "@referredType": "Individual"
      }
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "state": "cancelled",
      "@type": "PortabilityOrderItem",
      "action": "port",
      "portabilityActivationMode": "automatic",
      "portabilityEffectiveDate": "2025-01-22T08:00:00.000Z",
      "endUserName": "Lisa Anderson",
      "resource": {
        "@type": "LogicalResource",
        "@baseType": "Resource",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventory/v5/resource/987659",
```


© TM Forum 2026. All Rights Reserved.

```
        "id": "987659",
        "name": "+61412345684",
        "description": "Mobile telephone number - port cancelled",
        "category": "TelephoneNumber",
        "resourceCharacteristic": [
          {
            "@type": "StringCharacteristic",
            "@baseType": "Characteristic",
            "id": "1",
            "name": "NumberType",
            "value": "Mobile",
            "valueType": "string"
          }
        ],
        "resourceSpecification": {
          "@type": "ResourceSpecificationRef",
          "href": "https://mycsp.com:8080/tmf-
api/resourceCatalog/v5/resourceSpecification/12345",
          "id": "12345",
          "name": "Mobile Number Specification",
          "@referredType": "ResourceSpecification",
          "version": "1.0"
        }
      }
    }
  ],
  "donor": {
    "@type": "ServiceProvider",
    "href": "https://mycsp.com:8080/tmf-api/resourcePortability/v5/partyRole/5646",
    "id": "5646",
    "name": "Notel Communications Service Provider",
    "role": "ServiceProvider",
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/organization/1234",
      "id": "1234",
      "name": "Notel Communications Inc",
      "@referredType": "Organization"
    }
  }
}
```


Deletes a portabilityOrder

`DELETE /portabilityOrder/{id}?{filtering}`

Description

This operation deletes a PortabilityOrder entity.

Usage samples

Delete a portability order by its unique identifier

Request

© TM Forum 2026. All Rights Reserved.

```
DELETE /portabilityOrder/12345
Content-Type: application/json
```

Response

```
204
```

Operations on CancelPortabilityOrder

Retrieves a cancelPortabilityOrder by ID

`GET /cancelPortabilityOrder/{id}?fields=…`

Description

This operation retrieves a CancelPortabilityOrder entity. Attribute selection is enabled for all first level attributes.

Usage samples

Retrieve a specific cancel portability order request using its unique identifier

Request

```
GET /cancelPortabilityOrder/CANCEL-2024-001
Content-Type: application/json
```

Response

```
200

{
  "id": "CANCEL-2024-001",
  "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CANCEL-2024-001",
  "@type": "CancelPortabilityOrder",
  "requestedCancellationDate": "2025-01-22T09:00:00.000Z",
  "effectiveCancellationDate": "2025-01-22T09:30:00.000Z",
  "cancellationReason": "Technical issue with donor provider system",
  "state": "done",
  "portabilityOrder": {
    "@type": "PortabilityOrderRef",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/997",
    "id": "997",
    "name": "Portability order for +61412345676",
    "@referredType": "PortabilityOrder"
```

© TM Forum 2026. All Rights Reserved.

```
  }
}
```

List or find cancelPortabilityOrder objects

`GET /cancelPortabilityOrder?fields=…&{filtering}`

Description

This operation list CancelPortabilityOrder entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of cancel portability order requests with date range filtering, sorting, and pagination

Request

```
GET /cancelPortabilityOrder?offset=0&limit=20&before=2025-02-01T00:00:00Z&after=2025-
01-01T00:00:00Z&sort=requestDate:desc&filter=state=acknowledged
Content-Type: application/json
```

Response

```
200

[
  {
    "id": "CPO-001",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CPO-001",
    "@type": "CancelPortabilityOrder",
    "requestedCancellationDate": "2025-01-21T14:30:00.000Z",
    "effectiveCancellationDate": "2025-01-21T15:00:00.000Z",
    "cancellationReason": "Customer request - service no longer required",
    "state": "done",
    "portabilityOrder": {
      "@type": "PortabilityOrderRef",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/999",
      "id": "999",
      "@referredType": "PortabilityOrder"
    }
  },
  {
    "id": "CPO-002",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CPO-002",
```

© TM Forum 2026. All Rights Reserved.

```
    "@type": "CancelPortabilityOrder",
    "requestedCancellationDate": "2025-01-21T16:00:00.000Z",
    "effectiveCancellationDate": "2025-01-21T16:15:00.000Z",
    "cancellationReason": "Duplicate order created by mistake",
    "state": "done",
    "portabilityOrder": {
      "@type": "PortabilityOrderRef",
      "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/998",
      "id": "998",
      "@referredType": "PortabilityOrder"
    }
  }
]
```

Creates a cancelPortabilityOrder

`POST /cancelPortabilityOrder?fields=…&{filtering}`

Description

This operation creates a CancelPortabilityOrder entity.

Mandatory Attributes

Mandatory Attributes	Rule
portabilityOrder	
portabilityOrder.id	
@type	

Usage samples

Create a new cancellation request for an existing portability order

Request

```
POST /cancelPortabilityOrder
Content-Type: application/json

{
  "@type": "CancelPortabilityOrder",
  "requestedCancellationDate": "2025-01-21T16:00:00.000Z",
  "cancellationReason": "Duplicate order created by mistake",
  "portabilityOrder": {
    "@type": "PortabilityOrderRef",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/998",
    "id": "998",
    "@referredType": "PortabilityOrder"
  }
```

© TM Forum 2026. All Rights Reserved.

```
}
```

Response

```
201

{
  "id": "CPO-002",
  "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CPO-002",
  "@type": "CancelPortabilityOrder",
  "requestedCancellationDate": "2025-01-21T16:00:00.000Z",
  "cancellationReason": "Duplicate order created by mistake",
  "state": "acknowledged",
  "portabilityOrder": {
    "@type": "PortabilityOrderRef",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/998",
    "id": "998",
    "name": "Portability order for +61412345677",
    "@referredType": "PortabilityOrder"
  }
}
```

Updates partially a cancelPortabilityOrder

`PATCH /cancelPortabilityOrder/{id}?fields=…`

Description

This operation allows partial updates of a CancelPortabilityOrder entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
state	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable

© TM Forum 2026. All Rights Reserved.

Non Patchable Attributes	Rule
@type	@type is immutable

Patchable Attributes	Rule
cancellationReason	
effectiveCancellationDate	
portabilityOrder	
requestedCancellationDate	

Usage samples

Update a cancel portability order request using RFC 7396 JSON Merge Patch for simple field updates

Request

```
PATCH /cancelPortabilityOrder/CANCEL-2024-001
Content-Type: application/merge-patch+json

{
  "@type": "CancelPortabilityOrder",
  "portabilityOrder": {
    "id": "PO123456789",
    "@type": "PortabilityOrderRef"
  },
  "requestedCancellationDate": "2025-01-22T15:00:00.000Z"
}
```

Response

```
200

{
  "id": "CANCEL-2024-001",
  "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CANCEL-2024-001",
  "@type": "CancelPortabilityOrder",
  "requestedCancellationDate": "2025-01-22T15:00:00.000Z",
  "effectiveCancellationDate": "2025-01-22T14:30:00.000Z",
  "cancellationReason": "Technical issue with donor provider system",
  "state": "inProgress",
  "portabilityOrder": {
    "@type": "PortabilityOrderRef",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/997",
    "id": "997",
    "name": "Portability order for +61412345676",
```

© TM Forum 2026. All Rights Reserved.

```
    "@referredType": "PortabilityOrder"
  }
}
```

Update a cancel portability order request using JSON Patch with query capabilities for complex updates

Request

```
PATCH /cancelPortabilityOrder/CANCEL-2024-002
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/cancellationReason",
    "value": "Cancellation request rejected - port already completed"
  },
  {
    "op": "add",
    "path": "/note/-",
    "value": {
      "@type": "Note",
      "author": "System",
      "date": "2025-01-23T10:00:00.000Z",
      "text": "Cancellation rejected as port has already been completed"
    }
  }
]
```

Response

```
200

{
  "id": "CANCEL-2024-002",
  "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CANCEL-2024-002",
  "@type": "CancelPortabilityOrder",
  "requestedCancellationDate": "2025-01-23T09:00:00.000Z",
  "cancellationReason": "Cancellation request rejected - port already completed",
  "state": "rejected",
  "portabilityOrder": {
    "@type": "PortabilityOrderRef",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/998",
    "id": "998",
    "name": "Portability order for +61412345677",
    "@referredType": "PortabilityOrder"
  }
}
```

© TM Forum 2026. All Rights Reserved.

Update a cancel portability order request using RFC 6902 JSON Patch operations (add, remove, replace)

Request

```
PATCH /cancelPortabilityOrder/CANCEL-2024-003
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/effectiveCancellationDate",
    "value": "2025-01-24T11:45:00.000Z"
  },
  {
    "op": "add",
    "path": "/note/-",
    "value": {
      "@type": "Note",
      "author": "System",
      "date": "2025-01-24T11:45:00.000Z",
      "text": "Portability order cancellation completed successfully"
    }
  }
]
```

Response

```
200

{
  "id": "CANCEL-2024-003",
  "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CANCEL-2024-003",
  "@type": "CancelPortabilityOrder",
  "requestedCancellationDate": "2025-01-24T10:00:00.000Z",
  "effectiveCancellationDate": "2025-01-24T11:45:00.000Z",
  "cancellationReason": "Customer changed service provider preference",
  "state": "done",
  "portabilityOrder": {
    "@type": "PortabilityOrderRef",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/999",
    "id": "999",
    "name": "Portability order for +61412345678",
    "@referredType": "PortabilityOrder"
  }
}
```

Partially update an existing cancel portability order request using standard JSON merge patch

© TM Forum 2026. All Rights Reserved.

Request

```
PATCH /cancelPortabilityOrder/CANCEL-2024-004
Content-Type: application/json

{
  "@type": "CancelPortabilityOrder"
}
```

Response

```
200

{
  "id": "CANCEL-2024-004",
  "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/cancelPortabilityOrder/CANCEL-2024-004",
  "@type": "CancelPortabilityOrder",
  "requestedCancellationDate": "2025-01-25T08:00:00.000Z",
  "cancellationReason": "Business decision to retain current provider",
  "state": "acknowledged",
  "portabilityOrder": {
    "@type": "PortabilityOrderRef",
    "href": "https://mycsp.com:8080/tmf-
api/resourcePortability/v5/portabilityOrder/1000",
    "id": "1000",
    "name": "Portability order for +61412345679",
    "@referredType": "PortabilityOrder"
  }
}
```

Deletes a cancelPortabilityOrder

`DELETE /cancelPortabilityOrder/{id}?{filtering}`

Description

This operation deletes a CancelPortabilityOrder entity.

Usage samples

Delete a cancel portability order request by its unique identifier

Request

```
DELETE /cancelPortabilityOrder/12345
Content-Type: application/json
```

Response

© TM Forum 2026. All Rights Reserved.

```
204
```

© TM Forum 2026. All Rights Reserved.

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42
```

© TM Forum 2026. All Rights Reserved.

```
{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

© TM Forum 2026. All Rights Reserved.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

© TM Forum 2026. All Rights Reserved.

Acknowledgements

Release History

Release Number	Date	Release led by:	Description
5.0.0	20-Jun-2026	Abhilash Prasad, Jio	V4 to V5 Migration with comprehensive schema refactoring, event notification alignment, and enhanced documentation. Includes post-review updates: ServiceProvider schema relocation, TMF669 Party Role API relationship, and sequence diagram corrections

Contributors to Document

Version No	Date	Modified By	Description

© TM Forum 2026. All Rights Reserved.

5.0.0	20-Jun-2026	Abhilash Prasad, Jio	V4 to V5 Migration including: Schema reorganization (moved from Portability to Resource folder), ServiceProvider refactored as PartyRole (located in EngagedParty folder), PortabilityOrder updated with donor/recipient structure, PortabilitySubject enhanced with Authorization entity, PortabilityOrderItem consolidated and improved, Event notifications aligned with TMF v5 patterns (stateChange, informationRequired), TMF669 Party Role API relationship documentation, Sequence diagram showing API as interface exposed by service provider systems (not as standalone organizational entity), Complete example updates, Enhanced documentation with sequence diagrams, Rules and conformance profile updates

© TM Forum 2026. All Rights Reserved.

© TM Forum 2026. All Rights Reserved.